



Bachelor Thesis

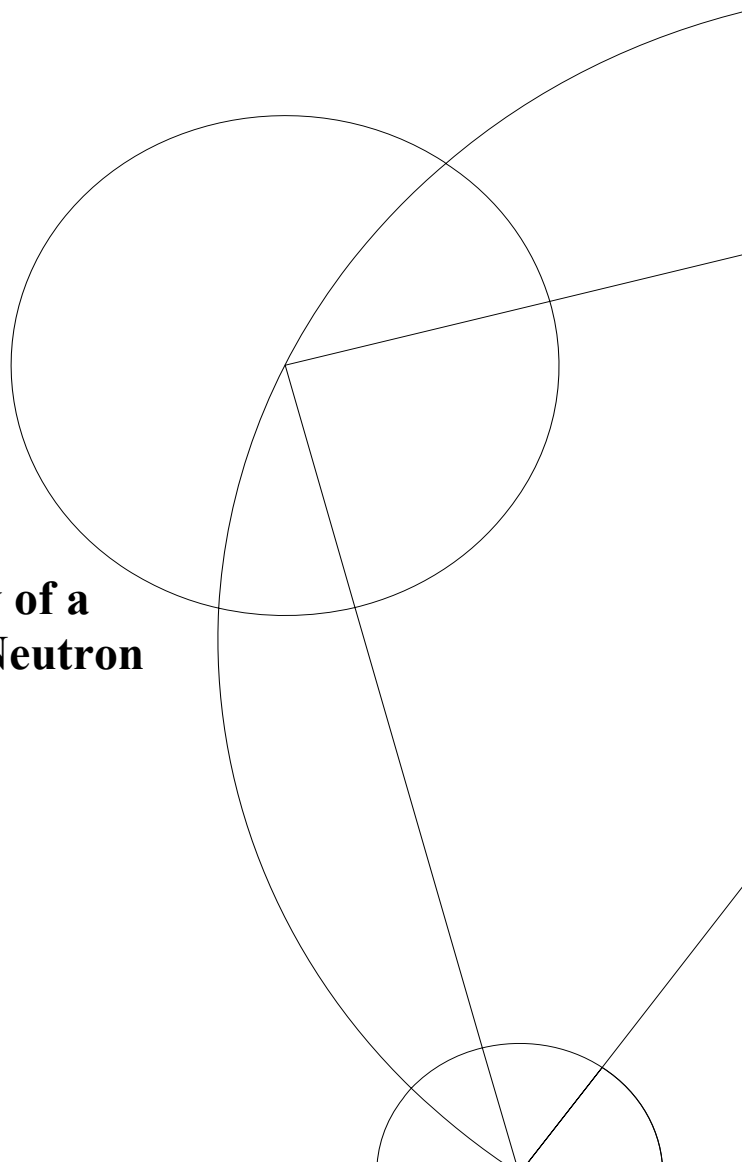
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Investigating the Optimal Geometry of a Vanadium Calibration Sample for Neutron Scattering

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Abstract

This report has created an independent minimalist neutron ray tracing software, McMini, which can simulate multiple scattering within cylindrical and spherical samples. McMini validates the simulation results obtained using the Union implementation in McStas, down to statistical noise. The agreement was confirmed up to neutrons with a scatter count of 1, 2...5

Identical simulation instruments of downstream sources on a spherical and cylindrical monitor have uncovered some monitor issues within McStas. We have shown that a north-to-south asymmetry is present, where intensity is biased towards the northern hemisphere in the spherical monitor. The cylindrical monitor records zero intensity in the highest angular bin column, and all data have been shifted to the left by one bin, thus giving the cylinder monitor a left-to-right asymmetry.

The report has found several valid geometries for the hollow vanadium sampled through multiple objective optimization. The multiple objective optimization located the Pareto solution set by minimizing a flatness metric and maximizing the mean intensity within a 60° detector window.

Using the Born approximation, we obtain a naive estimate of the scattered intensity from a vanadium sample. By comparing the expected intensity scattered to the observed intensity, we find a self-absorption ratio. For the PSI vanadium sampled that value is 0.4

Contents

Abstract	ii
1 Introduction	1
1.1 Project objectives	1
2 Theoretical background	1
2.1 Scattering and absorption cross sections	2
2.2 Neutron transport and Monte Carlo method	2
2.3 Simulating neutron scattering	3
3 McMini methodology and validation of Union in McStas	3
3.1 Design philosophy and purpose	3
3.2 Source_simple model and implementation	3
3.3 4PI monitor implementation	4
3.3.1 Comparison with McStas 4π monitor	5
3.4 Cylindrical monitor bin offset	6
3.5 Sample implementation	7
3.6 Validation	8
3.6.1 What is matched, metrics, and statistical treatment	8
3.6.2 Solid cylinder validation	8
3.6.3 Summary of validation results	11
4 Scattering Quality and Geometry Optimization in McStas	12
4.1 Simulation Setup in McStas	12
4.2 Scattering Quality Metrics	13
4.2.1 Deviation from expected single-scattering intensity	15
4.3 Full Instrument Test with PSI Geometry	16
4.4 Optimization Study and Recommended Geometry for Calibration Use	18
4.4.1 Objective of the Optimization study	18
4.4.2 Pareto optimization	18
4.4.3 Recommended Geometry for Calibration Use	18
5 Discussion	20
5.1 Physical interpretation of trends	20
5.2 Limitations of Monte Carlo model	21
5.3 Statistical precision	21
5.4 Limits of Union	21
6 Conclusion	21
6.1 Validation summary	21
6.2 Suggested Geometries and Pareto Optimization Results	21
6.3 Future work	22
A Theoretical background derivations	24
A.1 Derivation of Path Length Sampling	24
A.2 Derivation of the expected per bin intensity	25

B	Detailed Technical Derivations and Geometric Proofs for McMini	26
B.1	Derivation of flat monitor intersection and binning	26
C	Supplementary Validation Data: Solid Vanadium Sphere	27
C.1	Spatial intensity comparison	27
C.2	Scatter-order analysis	30
D	Technical Note on the 4π Monitor Asymmetry	32
D.1	Structural analysis on the cause of the observed asymmetry	32
D.2	Asymmetry analysis on the flat monitor vs. spherical monitor	33
D.2.1	Flat monitor	33
D.2.2	Spherical monitor	35

1 Introduction

Neutron scattering experiments are useful for determining the dynamics and structure of various materials. Neutrons interact with atomic nuclei via the strong nuclear force and magnetic moments via the electromagnetic force, rather than the electron cloud. This means that it can differentiate even light isotopes [1, p. 7]. Neutron simulation tools are useful for preparing for real experiments at spallation sources. Having a pre-prepared experiment already simulated increases efficiency in design. The reliability of these tools are therefore crucial, and being confident in these tools to correctly apply the physical properties will help designing an improved calibration sample.

Vanadium is widely used in the calibration process because its scattering is dominated by incoherent processes. The key property is its relatively large incoherent scattering cross section and relatively small coherent scattering cross section, which leads to an approximately isotropic scattering distribution. However, even for vanadium with isotropic scattering, the measured angular distribution will be influenced by the geometry of the sample. Accurate calibration samples are needed to determine detector efficiencies, instrument resolution, and systematic effects arising from sample environment, thus motivating the need to understand the geometry dependent scattering patterns.

McStas is a neutron ray tracing software commonly used to model neutron scattering experiments. McStas allows for very detailed descriptions of neutron sources, optical components, detectors, and samples. Complex geometries and multiple scattering events within composite samples can be simulated within McStas using the Union framework [2]. Rather than modeling scattering components individually, multiple geometric objects can be grouped into a single physical sample with Union. This allows neutrons to scatter multiple times within a single object before exiting [3]. The contribution of multiple scattering to measured intensities can be important, so it is critical that its implementation is physically correct. Therefore, independent verification of the multiple scattering behavior is useful. Verification against a minimal, stand-alone Monte Carlo implementation is performed within this thesis.

1.1 Project objectives

To begin with, this thesis presents the implementation of a minimalistic Monte Carlo neutron transport program written in the C programming language. This code was developed with the intention of creating a reference implementation to independently verify that McStas exhibits expected behavior when using the Union multiple scattering framework.

Secondly, we intend to show agreement between our independent simulation code and McStas using simple geometries such as cylinders. Angular intensity distributions as well as scattering statistics will be compared between the two implementations.

Lastly, once the simulation framework has been validated, further simulations are performed in McStas. This program will then be used to analyze the anisotropic scattering caused by hollow cylinders made of vanadium. The objective is to find geometry that reduces systematic errors to produce useful calibration behavior for neutron scattering experiments like those used at PSI in Switzerland.

2 Theoretical background

This chapter will introduce and discuss the theoretical concepts required to understand the simulation methods and physical processes studied in this thesis. In this chapter, we will review the theoretical basis needed to understand neutron scattering. Fundamental concepts like microscopic cross sections and macroscopic cross sections will be reviewed, as well as the process of multiple scattering.

2.1 Scattering and absorption cross sections

The neutron scattering cross sections determine how well a system scatters neutrons. It is expressed by

$$\sigma = \frac{1}{\Psi} \frac{dN}{dt} \quad (2.1)$$

where σ has units of area, and dN/dt , units of $[n/s]$. The scattered intensity is normalized by the flux, so that σ is an intrinsic property of the system. at the single nucleus level σ can be understood as the effective area that is perpendicular to the beam. It is also referred to as the microscopic scattering cross sections [1, p. 12]. For experiments there are usually a lot of atoms, so we convert the microscopic cross sections into macroscopic cross sections. This can be accomplished by multiplying the number density of atoms in the material by the microscopic cross section [1, p. 13].

$$\mu_t = n\sigma_t \quad (2.2)$$

Where n is the density of the number of atoms. This macroscopic cross section represents the probability of interaction per unit length. Following rules of probability, The total macroscopic cross section is additive leading to [1, p. 30].

$$\Sigma_t = \Sigma_s + \Sigma_a, \quad (2.3)$$

where Σ_s and Σ_a are the macroscopic scattering and absorption cross sections, respectively [1, p. 30]. Scattering lengths vary, but since we are dealing with macroscopic samples, the sample can average over all configurations (ensemble average). If we write the scattering length of nucleus j as

$$b_j = \langle b \rangle + \delta b_j, \quad (2.4)$$

then the coherent part of the scattering will be determined by the average scattering length $\langle b \rangle$, while the incoherent part is determined by the variance of the fluctuations. The total single nucleus scattering cross section is proportional to $\langle b^2 \rangle$ while the coherent contribution is proportional to $\langle b \rangle^2$. The difference is then incoherent.

$$\sigma_{\text{inc}} = 4\pi (\langle b^2 \rangle - \langle b \rangle^2) = 4\pi \langle (\delta b)^2 \rangle. \quad (2.5)$$

In the case of vanadium, the incoherent cross section is dominant with the tabulated value [1, p. 16]

$$\sigma_{\text{inc}} = 5.08 \text{ barn}. \quad (2.6)$$

This is much larger than the coherent value that has a value of $\sigma_{\text{coh}} = 4\pi b^2 = 0.0247$ barns. Since the incoherent cross section is dominate vanadium has isotropic scattering with very little Bragg peaks. The absorption cross section depends on neutron velocity which will be important to model correctly in the simulation. The absorption cross section follows an inversely proportional relationship with velocity [1, p. 15].

2.2 Neutron transport and Monte Carlo method

The key concepts in Monte Carlo simulation is the sampling of free-flight distances using random numbers. The Free-flight distances s represent the distance between interactions events. The probability that a neutron travels a distance s without interacting follows the Lambert-Beer law for attenuation [1, p. 31].

$$P(s) = e^{-\Sigma_t \cdot s}, \quad (2.7)$$

where Σ_t is the total macroscopic cross section of the material. Free-flight distances are therefore sampled from the distribution [4]

$$s = -\frac{\ln(U)}{\Sigma_t} \quad (2.8)$$

Where U is a uniformly distributed random number between $[0, 1]$. Random numbers are then again used to decide whether an event is a scattering or absorption event. By applying this neutron transport logic for a large number of neutrons, statistical estimates of measurable quantities, such as intensity distribution, can be obtained.

2.3 Simulating neutron scattering

In neutron scattering experiments it is possible for the neutron to scatter multiple times before leaving the sample. These intermediate scattering events may deflect the neutron, transfer energy to the neutron, or absorb the neutron, altering the observed distribution. Multiple scattering can strongly impact the resulting signal, depending on sample thicknesses of arbitrary geometries, or from samples with large cross sections [1, p. 7]. Calibration samples like vanadium have a relatively large scattering cross section [1, p. 16] and could experience multiple scattering, altering the final scatter distribution and deviating from an ideal isotropic pattern [5]. In order to model neutron scattering, one must keep track of the neutron through sequential histories through the sample until it exits or is absorbed. This process lends itself to Monte Carlo simulations, since neutron histories can be tracked through individual interactions, where the location of the next collision can be sampled from Lambert-Beer law for attenuation [6, 4]. In Monte Carlo neutron capabilities, it becomes useful to assign weight to neutrons since portions of the code use biasing or restricted sampling. The source may only target a finite area instead of sampling uniformly across all angles. The ray weight must account for the proper phase-space factor so that ray intensity can be properly normalized. p thus represents the physical intensity of that neutron history.

3 McMini methodology and validation of Union in McStas

3.1 Design philosophy and purpose

McMini is an independently developed minimal Monte Carlo neutron transport system implemented in C. McMini is simple by design, with the philosophy of reducing primary physical interactions down to its core elements to best recreate the results of a real experiment. Its purpose is to validate and reference the result of the multiple scattering experiment performed with Union in McStas.

3.2 Source_simple model and implementation

The source implemented in McMini follows the same input parameters as Source_simple in McStas [7]. In order to compare the two ray tracing programs, they must match in the neutron generation. For the source, the spatial, directional, wavelength, and weight distributions of the McMini source must match those of McStas. A neutron ray is described in the system by the trajectory

$$\vec{r}(t) = \vec{r}_0 + \vec{v}t, \quad (3.1)$$

where $\vec{r}_0$ marks the ray's origin and $\vec{v}$ denotes its directional vector. Since our model ignores gravity and time of flight effects, we allow t to act as a measure of path length instead of physical time. Here, $\vec{r}_0$ is drawn from a uniform distribution on a circular source disk. The direction is sampled by choosing a point in the rectangular

target area and normalizing the vector from the source point to the sampled target point. The wavelength is sampled uniformly according to

$$\lambda \sim \mathcal{U}[\lambda_0 - \Delta\lambda, \lambda_0 + \Delta\lambda]. \quad (3.2)$$

As described in Section 2.3, each ray carries a Monte Carlo weight p . In the McMini implementation of `Source_simple`, the source weight is factorized as

$$p = p_{\text{mul}} \cdot p_{\text{dir}}, \quad (3.3)$$

where

$$p_{\text{mul}} = \Psi 10^4 \pi R^2 2\Delta\lambda \quad (3.4)$$

and

$$p_{\text{dir}} = \frac{x_{\text{focus}} y_{\text{focus}}}{d^2} \cos^2 \theta. \quad (3.5)$$

Here Ψ is the source flux, R is the radius of the circular source, x_{focus} and y_{focus} are respectively the horizontal and vertical lengths of the rectangular target area, d [m] is the distance between the sampled source point and the sampled target point, and θ is the angle between the ray's direction and normal of the source. The factor 10^4 is present to account for unit conversion in the definition of flux from cm^{-2} to m^{-2} . As can be seen from the above definitions, unlike p_{mul} , p_{dir} depends on the position sampled on the source and target areas since both d and θ will vary for each ray.

3.3 4PI monitor implementation

The 4PI monitor is a sphere with radius R . It has n_x bins along a latitude line and n_y bins along a longitude line. The spherical monitor is the obvious choice when investigating how neutrons scatter in all directions. The intersections of rays with the sphere are found by geometrically describing the sphere with a quadratic equation and solving the equation for the t -values. The hit points are found by solving the quadratic equation

$$(x_0 + v_x t)^2 + (y_0 + v_y t)^2 + (z_0 + v_z t)^2 = R^2 \quad (3.6)$$

where after solving for t the hit point is computed using

$$\vec{r}(t) = \vec{r}_0 + \vec{v} \cdot t \quad (3.7)$$

We compute ϕ and θ are found by taking the unit vector of the hit position to the center of the 4π monitor, and converting it to angular coordinates.

$$\phi = \arctan \frac{u_x}{u_z}, \quad \phi \in [-180, 180] \quad (3.8)$$

$$\theta = \arctan \frac{u_y}{\sqrt{u_x^2 + u_z^2}}, \quad \theta \in [-90, 90] \quad (3.9)$$

The hit points are subsequently binned using the normalized bin indices

$$f_\phi = \frac{\phi - \phi_{\min}}{\phi_{\max} - \phi_{\min}}, \quad f_\theta = \frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}} \quad (3.10)$$

After that, we compute which bin the hit belongs in and increment the bin count.

$$i_x = \lfloor f_\phi n_x \rfloor, \quad i_y = \lfloor f_\theta n_x \rfloor \quad (3.11)$$

Finally, the intensity per pixel that is reported is

$$I_{ij} = \frac{1}{N_{\text{count}}} \sum_{\text{hits}} p_{\text{mul}} p_{\text{dir}}. \quad (3.12)$$

Using p_{mul} and p_{dir} as discussed in Section 3.2, i, j denotes the corresponding pixels and N_{count} denotes the number of neutrons. No geometrical solid-angle normalization is applied internally. Thus, the raw monitor signal corresponds to the weighted sum of neutron probabilities per pixel.

Longitude vs latitude binning. The Longitude ϕ and latitude θ bins are defined on the 4PI monitor. This convention enables us to plot the data on a rectangular map. This is convenient, however, equal steps in latitude and longitude are not equal in solid angle. In particular, the bins near the poles have a smaller solid angle than those near the equator. We therefore see that the rectangular map does not directly represent the intensity per solid-angle element. The differential solid angle when θ is the latitude angle

$$d\Omega = \cos \theta d\theta d\phi. \quad (3.13)$$

So, the solid angle between latitudes θ_1 and θ_2 is [8]

$$\Omega = \int_0^{2\pi} \int_{\theta_1}^{\theta_2} \cos \theta d\theta d\phi = 2\pi (\sin \theta_2 - \sin \theta_1). \quad (3.14)$$

As a result, the bins located at the poles represent smaller areas. Therefore, we create an array of size $[n_x, n_y]$ where we take $\cos \theta$ at that latitude. Each bin was divided by $\cos \theta$ where θ is the latitude of that bin, effectively converting the intensity to solid-angle elements.

3.3.1 Comparison with McStas 4π monitor

To validate the 4π monitor in McMini and McStas the same settings were applied to a downstream simulation that contained only the source and the monitor, allowing a direct comparison between the two.

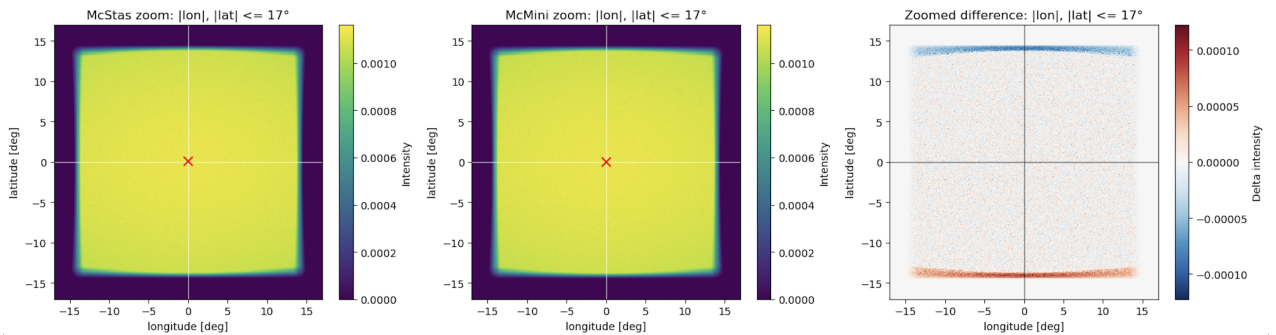


Figure 3.1: Zoomed comparison and difference map of the direct beam on a 4π monitor.

The comparison plot can be seen in Figure ?? using the same color scale. Visually, the two direct beam distributions appear identical. However, if one subtracts the McStas data from the McMini data (Figure ??),

there appears to be a systematic difference larger than statistical noise. The difference lies close to the top and bottom edge of the direct beam footprint on the 4π monitor, not along the edge of the entire 4π monitor. The edge of concern is then the $\sim 15^\circ$ that the direct beam takes up on the 4π monitor. To test if this difference was due to a north to south asymmetry, the southern hemisphere was mirrored over to the northern hemisphere. This comparison was done for both McStas and McMini as seen in Figure 3.2.

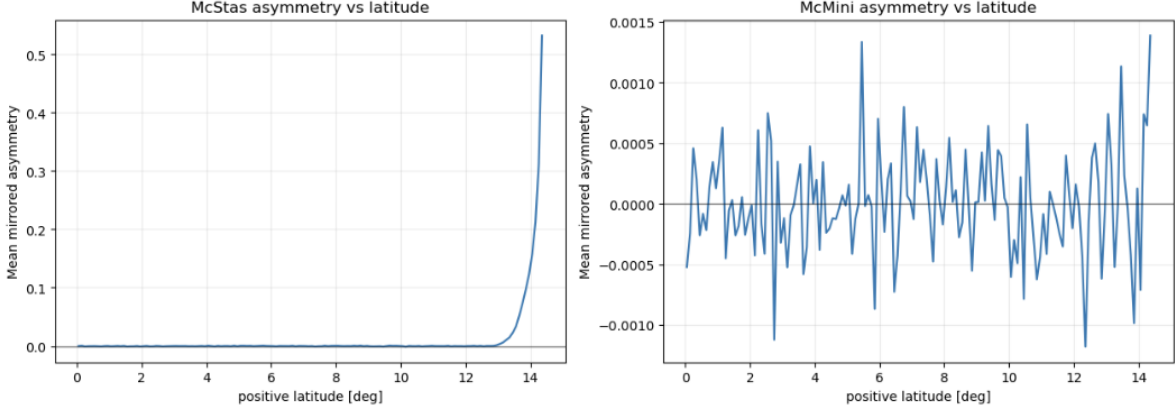


Figure 3.2: Mirrored direct beam map and north to south asymmetry as a function of latitude.

Figure 3.2 shows that the McStas 4π monitor contains a systematic north to south asymmetry near the edge of the direct beam footprint. The asymmetry increases towards the angular boundary of the beam approximately 14° . In contrast, the corresponding McMini result shows no systematic north to south asymmetry, and the remaining difference is consistent with statistical noise. Interestingly, the issue was not present when testing the same setup on a flat plane monitor. Computing the degree of asymmetry with

$$A = \frac{|I_N - I_S|}{I_N + I_S}, \quad (3.15)$$

we see McStas shows a north to south asymmetry of 0.37%. The observed north-south asymmetry was investigated in detail. For the complete structural analysis of this asymmetry, see the Appendix D. With the knowledge of this asymmetry we move on.

3.4 Cylindrical monitor bin offset

A cylindrical monitor was implemented, and in the comparison with McStas' Cyl_monitor_PSD three issues were discovered. The rightmost angular bin column records 0 intensity. There is a left right asymmetry. The top and bottom row has an unexplained drop in intensity. The two issues are correlated with each other. The left-right asymmetry is likely caused by an angular bin shift and thus the rightmost bin column missing. A downstream beam onto the cylindrical monitor showed that the normalized left right asymmetry is

$$\frac{|I_L - I_R|}{I_L + I_R} = 3.60 \times 10^{-2}. \quad (3.16)$$

Shifting the monitor data one angular bin to the right ($\Delta b = +1$) and dropping the empty left edge bin restores symmetry:

$$\frac{|I_L - I_R|}{I_L + I_R} = 1.63 \times 10^{-6}. \quad (3.17)$$

Neither of the adjacent shifts correctly realigns the beam:

$$\Delta b = -1 : 5.12 \times 10^{-2}, \quad \Delta b = +2 : 3.60 \times 10^{-2}. \quad (3.18)$$

Therefore, $\Delta b = +1$ is uniquely the amount needed to minimize the beam asymmetry, reducing its magnitude by effectively 100%. Thus, it seems reasonable to conclude that the cylindrical monitor suffers from a one bin angular displacement.

3.5 Sample implementation

Intersection logic

Finding the intersections of the cylinder is slightly different to that of the sphere. Finding the intersections is not alike that of computing intersection with the sphere monitor. The cylinder is slightly more complicated as it has finite height along the y-axis. The code finds possible intersections and used two conditions, which are combined to compute the entry and exit point. Finding the intersection with the walls and lids required solving these two equations respectively

$$(x_0 + v_x t)^2 + (z_0 + v_z t)^2 = R^2 \quad \text{and} \quad \pm \frac{H}{2} = y_0 + v_y t. \quad (3.19)$$

Where R is the radius of the cylinder and H is the height of the cylinder. Solving the two equations provides a total of 4 t-values. Therefore, conditions are needed to select the correct t-values that represent the entry and exit point. Since the ray cannot exit the walls if it is above or below the lids and it cannot exit the lids if it is outside the wall, we obtain two conditions. Written out as the radial condition and the height condition:

$$x^2 + z^2 \leq R^2 \quad \text{and} \quad -\frac{H}{2} \leq y \leq \frac{H}{2} \quad (3.20)$$

Using the two conditions, the correct entry and exit point is picked and used to compute the path length. From here the microscopic cross sections are converted to macroscopic cross sections, the absorption terms was scaled with $\frac{2200}{v}$ [1, p. 15], and the free flight distance was sampled.

Free flight distance sampling logic

The path length refers to how far the ray travels inside the material. The free-flight distance refers to how far the ray travels before interaction. Free-flight distance is computed by uniformly sampling a random number in $[0, 1]$ and using

$$s = -\frac{\ln U}{\Sigma_t}. \quad (3.21)$$

If the free-flight distance is larger than the path length L , then the neutron is considered transmitted to the exit point of the surface. If the free-flight distance was shorter than the path length L , then an interaction happened with the material. The ray is moved to where the interaction took place. The code then determines if it was scattering or absorption by comparing a random number to the probability that it was scattering.

$$P_{\text{scat}} = \frac{\Sigma_{\text{scat}}}{\Sigma_t} \quad (3.22)$$

If the random number is larger than P_{scat} , it is killed and discarded. Otherwise, it is counted as a scattering event. The code then samples a new direction by sampling uniformly from the unit sphere. It picks a random azimuthal angle and cosine of the polar angle.

Multiple scattering algorithm

Transport logic includes two interfaces related by functionality. `scatterer_sphere_interact` only takes one ray and does a single interaction step. It will determine whether or not the ray missed the sphere, transmitted through the sphere, was absorbed by the sphere, or scattered inside the sphere. This function performs ray sphere intersection and calls the isotropic scattering function once per ray. `transport_sphere` calls `scatterer_sphere_interact` many times inside of a loop. It only stops when ray becomes dead inside of the sphere, either by being transmitted out or absorbed. Since this function calls `scatterer_sphere_interact` in a loop it allows the ray to multiple scatter while inside the sphere. It has a fail-safe maximum number of scattering events to prevent infinite loops. As long as the particle is alive it will call ray intersection function afterwards. When a ray is absorbed, it will be killed and removed from the tally. Unlike McStas which weights probability that ray will continue (reduces variance), McMini ray is simply discarded.

3.6 Validation

3.6.1 What is matched, metrics, and statistical treatment

To validate the Union framework in McStas, the simulated scattering patterns are compared with those obtained from the independently developed Monte Carlo program McMini. The comparison includes spatial intensity distributions, difference maps between the two simulations, integrated intensities, and scatter-order statistics. The difference map is defined as the bin-by-bin intensity difference between McStas and McMini, or simply as

$$\Delta I = I_{\text{Union}} - I_{\text{McMini}} \quad (3.23)$$

In addition to full spatial maps, equatorial cuts of the intensity distribution are analyzed in order to compare the angular dependence of the scattering signal along the detector longitude. Quantitative agreement between the simulations is evaluated using statistical measures such as the root-mean-square error (RMSE).

3.6.2 Solid cylinder validation

Simulation parameters

All parameters used in McStas are shown in table 3.1 and McMini settings were set to match. Parameters not mentioned can be assumed to be the default values of McStas [9]

Table 3.1: Parameters used in the McStas validation simulation for comparing the vanadium scattering pattern on the PSD_monitor_4PI.

Component	Parameters
Source_simple	$\lambda_0 = 4.0$, $\Delta\lambda = 0.1$, focus_xw= 0.0015, focus_yh= 0.025, dist= 1.0, radius= 0.01, $n = 10^9$
Incoherent_process	$\sigma = 5.08$, unit_cell_volume= 13.827
Union_make_material and Union_cylinder	my_absorption= 36.7, radius= 0.005, height= 0.02
PSD_monitor_4PI	nx= 360, ny= 180, radius= 1.0, restore_neutron= 1

Spatial intensity and difference comparison

To compensate for the solid angle, each bin was divided by the geometric factor $\cos \theta$ discussed in Section 3.3. The corrected maps are shown in Figure 3.3. The poles include a higher degree of noise, since the bins are smaller and thus receive fewer rays. Once the correction is applied, the noise can blow up. 5° of the poles have been removed for this reason as well as to reduce the asymmetry problem. Then subtracting the two maps gives the difference map in Figure 3.3

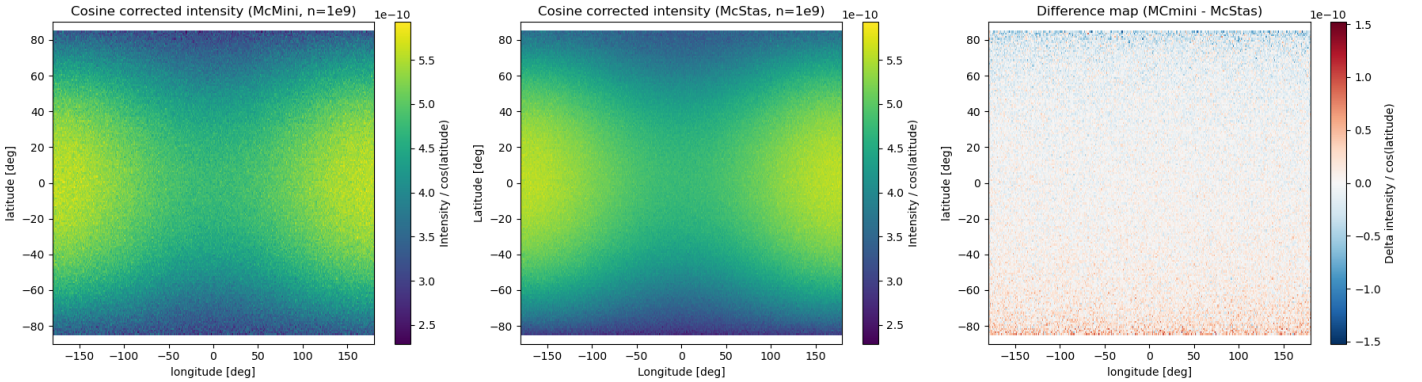


Figure 3.3: Corrected intensity map and difference map of a solid cylinder with a radius of 0.5cm and height of 2cm

The difference map shows mostly noise, but some clear bias is visible. A north bias, which is believed to come from the 4π monitor is visible. This is where the asymmetry was initially discovered. Quantifying the difference map, using mean residuals, RMS residuals, mean pull, and reduced chi-square, gives the values in Table 3.2

Metric	Cosine-corrected map
Mean residual	$-9.328943 \times 10^{-14}$
Mean absolute residual	1.399019×10^{-11}
RMS residual	1.867373×10^{-11}
Mean pull	-1.522858×10^{-2}
Pull standard deviation	1.094898
Reduced χ^2	1.199033

Table 3.2: Quantitative comparison of the cosine-corrected difference map between McMini and McStas for the solid cylinder case.

For the cosine corrected map, the mean residual ($-9.33 \cdot 10^{-14}$) shows agreement in total intensity, while the residual RMS (1.40×10^{-11}) shows a typical mismatch per bin of the order of 10^{-11} . The pull width was 1.09 and the reduced χ^2 was 1.20.

Equator and meridian cut

The equator cut is computed as the mean intensity in a 4° wide latitude band centered on the equator, $\theta \in [-2^\circ, 2^\circ]$, and is plotted as a function of longitude ϕ . The meridian cut is computed as the mean intensity in a 4° wide longitude band centered on the meridian, $\phi \in [-2^\circ, 2^\circ]$, and is plotted as a function of latitude θ . The equator cut is shown in the left panel of Figure 3.4, while the meridian cut is shown in the right panel.

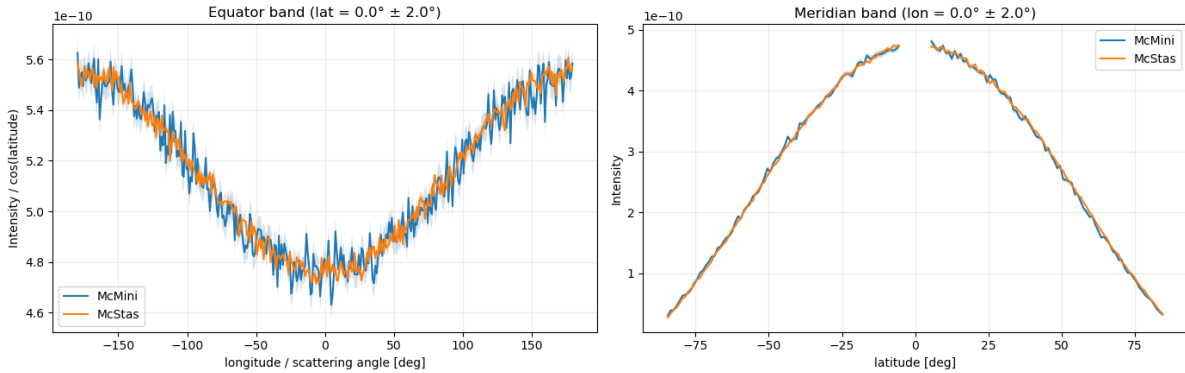


Figure 3.4: Equator and meridian cut for the solid cylinder case

The cuts show excellent agreement. McMini shows higher levels of noise. This is due to McStas performing variance reduction by adjusting for absorption by weight adjustment, rather than removing absorbed neutron rays, as in McMini.

Integrated intensities

After applying the 5° pole cut and cosine correction, the integrated intensities are

$$I_{\text{McMini}} = 2.766574 \times 10^{-5}, \quad I_{\text{McStas}} = 2.767145 \times 10^{-5}.$$

The ratio is

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} = 0.9998209,$$

Showing excellent agreement. Integrating the northern and southern hemispheres gives the intensities

	Northern Hemisphere	Southern Hemisphere	Asymmetry $A = \frac{N-S}{N+S}$
McMini	1.383150×10^{-5}	1.383425×10^{-5}	-9.944478×10^{-5}
McStas	1.404191×10^{-5}	1.362954×10^{-5}	1.490263×10^{-2}

Table 3.3: Integrated intensities over northern and southern hemispheres and corresponding asymmetries for McMini and McStas simulations.

Table 3.3 shows disagreement in the intensity recorded in the hemispheres of the 4π monitor and a clear asymmetry in the simulation of a solid vanadium cylinder recorded on the McStas 4π monitor. The validation performed on the solid vanadium cylinder was also performed on a solid vanadium sphere for extra validation. Interestingly, the asymmetry levels agree closely. For the solid vanadium sphere, the asymmetry levels recorded on the 4π monitor were $A_{\text{sphere}} = 1.429436 \times 10^{-2}$. For the validation of the solid vanadium sphere, see Appendix C

Scatter-order analysis

Last part of the validation will again contain the analysis of the maps with neutron of different scatter orders.

Order	I_{McMini}	Share _{McMini}	I_{McStas}	Share _{McStas}	$I_{\text{McMini}}/I_{\text{McStas}}$	RMS
1	2.419961×10^{-5}	87.47%	2.420378×10^{-5}	87.47%	0.9998278	1.729288×10^{-11}
2	3.016740×10^{-6}	10.90%	3.018256×10^{-6}	10.91%	0.9994977	5.013830×10^{-12}
3	3.904320×10^{-7}	1.41%	3.903869×10^{-7}	1.41%	1.0001150	1.720062×10^{-12}
4	5.118903×10^{-8}	0.19%	5.122508×10^{-8}	0.19%	0.9992964	6.133285×10^{-13}
5	6.732789×10^{-9}	0.02%	6.769408×10^{-9}	0.02%	0.9945905	2.229384×10^{-13}

Showing that the simulation agrees in total intensities, and they agree for neutrons of different scatter orders as well.

3.6.3 Summary of validation results

Agreement levels

This validation shows a very good agreement of these simulations between McMini and McStas. Visually, we can tell that the spatial intensity profiles overlap perfectly. The equator and meridian cuts show clearly they have the same angular dependence within increased statistical noise in McMini. The total integrated intensity agree within 10^{-4} , with ratios of

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} \approx 0.9998, \quad (3.24)$$

showing that identical normalization has been used in every case. Bin level agreement is also strong with residual RMS of 10^{-11} and reduced χ^2 values of ~ 1.2 , showing that the deviations arise mostly from noise. Scatter order decomposition agrees excellently, for neutrons of all scatter orders. Overall these results show that the Union implementation of McStas correctly reproduces the physical behavior of the simple neutron scattering program McMini.

Remaining discrepancies

The only systematic discrepancy observed is the north-south asymmetry present in McStas while McMini remained symmetric within statistical uncertainty.

$$A_{\text{McMini}} \sim 10^{-6},$$

McStas exhibits a consistent asymmetry of

$$A_{\text{McStas}} \approx 1.4 \times 10^{-2},$$

The asymmetry was observed in simulation of the solid vanadium cylinder discussed in Section 3.6.2 and in the extra validation of the solid vanadium sphere (Appendix C). This indicated that the effect is not geometry dependent. The asymmetry was visible in the reduced χ^2 as it was above unity. However, all cases showed the same total integrated intensity within statistical uncertainty, indicating a redistribution error. Diagnostics revealed that the problem originated in part from the component `PSD_monitor_4PI`. Tests indicated asymmetry from a downstream beam onto the monitor, while it did not show itself on a flat plane monitor. Apart from this observed discrepancy, all remaining differences can be attributed to statistical noise, with McMini having more noise, as it did not include the variance reduction method that McStas uses.

Confidence in Union implementations

The simplicity of McMini is crucial for confidence in the validity of the simulation results. McMini implements only the necessary physics, such as microscopic scattering and absorption length, unit cell volume, and wavelength dependent absorption. The propagation of rays through the simple geometries, and seeing that it matches McStas down to only a small north-south asymmetry, is a good result. McMini does not account for absorption by weight adjustment as in McStas and therefore presents a higher level of noise. This is actually a strength in the validation because it means that even when treating the absorption of neutron rays in a fundamentally different way, we still obtain the same result. This is great evidence that the Union framework correctly implements cross sections. The only difference observed between McMini and McStas that cannot be attributed to noise is the north-south asymmetry. The Union framework has been validated and is consistent with the physical models of cross sections. However, a binning or weight redistribution of the spherical monitor is required, and fixing the missing highest angular bin column in the cylindrical monitor.

4 Scattering Quality and Geometry Optimization in McStas

4.1 Simulation Setup in McStas

We were given a full instrument file from PSI that replicates the monitors, geometry, sample parameters, etc. The built in detectors, which were helium filled tubes, required a lot of computational power. They were replaced with a spherical monitor with the vanadium sample and the center and rotated so that the vanadium cylinder samples axis of rotation aligns with the poles. The simulation used these default parameter values unless anything else is specified.

Parameter	Value	Description
SourceE	5 meV	Mean source neutron energy
DeltaSourceE	0.5 meV	Width of source energy spectrum
EI	5 meV	Incoming neutron energy selected by the monochromator
A3	0°	Sample rotation angle
A4	−45°	Sample take-off angle / tank angle
SAMPLE	0	Sample type, where 0 corresponds to vanadium
RV_mono_Bool	1	Vertical monochromator focusing enabled
RH_mono_Bool	1	Horizontal monochromator focusing enabled
Lv	1.872 m	Vertical focusing distance
Lh	1.340 m	Horizontal focusing distance
monoShieldingWidth	0.062 m	Width of monochromator shielding opening
monoShieldingHeight	0.14 m	Height of monochromator shielding opening
monoShieldingDistance	0.235 m	Distance between monochromator and shielding opening
MonoShielding	1	Monochromator shielding enabled
Be_filter	1	Beryllium filter enabled
sampleHeight	0.01 m	Sample height
sampleRadius	0.01 m	Sample radius
sampleThickness	0.0026 m	Wall thickness of hollow cylindrical sample
samplePosY	0.00 m	Vertical offset of sample
sampleFluxDiagnosticXWidth	0.08 m	Width of diagnostic flux monitor
sampleFluxDiagnosticYHeight	0.08 m	Height of diagnostic flux monitor

Table 4.1: Relevant simulation parameters for the PSI CAMEA instrument setup.

The 4π monitor had a radius of **1m** and 180×90 angular bins. The monitor uses the `WHEN(raw_scattered > 0)` condition. This means that the monitor does not measure the direct beam, but only neutrons that have scattered at least once.

4.2 Scattering Quality Metrics

We assign key metrics to a simulation to quantify the quality of the sample. We will use these metrics in combination with each other and separately.

Summed intensity. The first metric is the total intensity of all bins within two symmetric windows centered on the equator ($\theta = 0$) and at longitude $\phi = \pm\phi_0$. The window has a longitude width of 60° and a latitude width of 4° chosen to match the detectors used in PSI. We define the sum as follows

$$I_{\text{tot}} = \sum_{(\phi_j, \theta_i) \in W} I(\phi_j, \theta_i). \quad (4.1)$$

where W are all bins within the two symmetric windows.

Reduced Chi-square. The second metric is the reduced chi-square (χ_ν^2) [10] obtained by fitting a constant, c , within the two windows. The constant c is fitted by taking the average of the latitude bins inside the window

and fitting to the longitude.

$$\chi^2_\nu = \frac{1}{N_\phi - 1} \sum_{j=1}^{N_\phi} \frac{(\bar{I}_j^{\text{band}} - c)^2}{\sigma_{\bar{I},j}^2}. \quad (4.2)$$

We lose one degree of freedom from fitting with one parameter. The mean intensity and variance was defined as

$$\bar{I}_j^{\text{band}} = \frac{1}{N_\theta} \sum_{|\theta_i| \leq \theta_0} I(\phi_j, \theta_i), \quad \sigma_{\bar{I},j}^2 = \frac{1}{N_\theta^2} \sum_{|\theta_i| \leq \theta_0} \sigma^2(\phi_j, \theta_i). \quad (4.3)$$

Coefficient of Variation. We will also look at the coefficient of variation to understand the relative variability of the mean intensity [11]. It is simply defined as

$$CV = \frac{\sigma_I}{\bar{I}}. \quad (4.4)$$

Expected single-scattering intensity. An important value is the self-absorption that is found from the observed intensity over the expected intensity. It comes from naively assuming only isotropic scattering and ignoring other effects such as multiple scattering and absorption. The naive estimate assumes that the neutrons reaching the sample scatter with probability P_{scatt} , and that the scattered intensity is then distributed isotropically over 4π . Start from the differential scattering cross section [1, p.13]. The expected intensity in one angular bin is then

$$I_{\text{exp}} = I_{\text{hit}} P_{\text{scatt}} \frac{\Delta\Omega}{4\pi}, \quad (4.5)$$

where I_{hit} is the intensity that hits the sample. I_{hit} is estimated by placing a monitor right before the sample which records all incoming neutron rays and a monitor right after which records all neutron rays that have not intersected the sample. Defining the intensity before and after by I_{before} and I_{after} we find I_{hit} by

$$I_{\text{hit}} = I_{\text{before}} - I_{\text{after}} \quad (4.6)$$

$\Delta\Omega$ is the solid angle of one detector bin. For our 2-by-2 degree bin that covers a solid angle of

$$\int_{\theta_1}^{\theta_2} \int_{-\phi_0}^{\phi_0} \cos \phi \, d\phi \, d\theta = 2 \sin \phi (\theta_2 - \theta_1) = 1.218 \cdot 10^{-3}. \quad (4.7)$$

We can estimate the probability of scattering from

$$P_{\text{scatt}} = \frac{\Sigma_{\text{inc}}}{\Sigma_t} (1 - \exp(-\Sigma_t \langle L \rangle)), \quad (4.8)$$

where $\langle L \rangle$ represents an estimated average distance traveled within the vanadium sample. $\Sigma_{\text{inc}} = 0.367 \text{ cm}^{-1}$ and $\Sigma_t = 0.734 \text{ cm}^{-1}$. Note that the projected area of the beam multiplied by the average path length gives what seems to be an effective volume being struck. Hence, we can write the average path length as volume divided by projected area $\langle L \rangle = \frac{V}{A_{\text{proj}}}$. For a thin sample, where self-absorption and multiple scattering are negligible, the ratio:

$$R = \frac{I_{\text{obs}}}{I_{\text{exp}}} \quad (4.9)$$

should approach one. Deviations from unity quantify neutron rays being self-absorbed or scattering multiple times.

4.2.1 Deviation from expected single-scattering intensity

We test the **expected intensity model** by approaching a very thin wall thickness. For very thin walls, few neutron rays should be absorbed and in the limit we should approach unity in the ratio. We used the PSI instrument file and default instrument parameters, except for the wall thickness. The uncertainties in the ratio were computed from the uncertainty in the observed and expected values. For the observed, the uncertainty was computed by adding the uncertainties of the observed bins in quadrature, and the error in the expected born estimate comes from the uncertainty of the incoming beam:

$$\sigma_O = \sqrt{\sum_i \sigma_{O_i}^2} \quad \text{and} \quad \sigma_E = E \frac{\sigma_{I_{\text{hit}}}}{I_{\text{hit}}}. \quad (4.10)$$

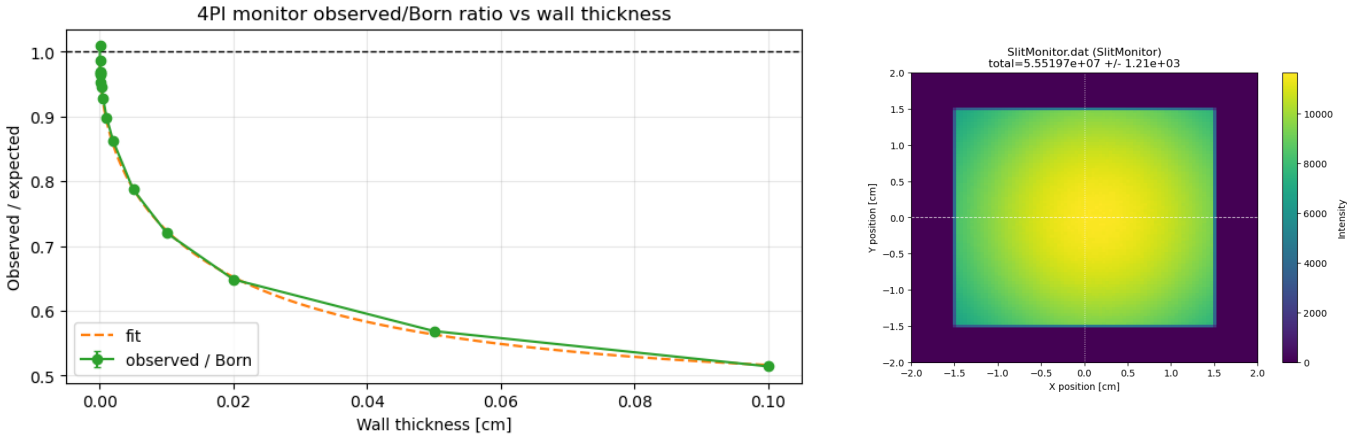
Propagating the uncertainty to the final ratio gives

$$\sigma_R = R \sqrt{\left(\frac{\sigma_O}{O}\right)^2 + \left(\frac{\sigma_E}{E}\right)^2}. \quad (4.11)$$

The key assumption was that the observed bin errors are uncorrelated, which is why they combine as $\sigma_O = \sqrt{\sum_i \sigma_{O_i}^2}$. The fitted model is a stretched exponential of the form

$$R(t) = c + A \cdot \exp(-(kt)^p) \quad (4.12)$$

chosen because it better captures how the tail falls off. A fit parameter p close to 1 means ordinary exponential decay with one characteristic thickness. We extrapolate the ratio at a thickness of 0 [cm] from $R(0) = c + A$. The simulations approaching the thin limit with the fitted stretched exponential are shown in Figure 4.1a



(a) Observed intensity over expected intensity as a function of wall thickness. (b) Incoming beam profile from slit monitor intensity distribution.

Figure 4.1: Comparison of the thin-limit intensity ratio and the slit monitor distribution.

The corresponding fit parameters are shown in Table 4.2. Since $p \neq 1$ the drop in the ratio does not simply follow one effective attenuation scale. The limit approaches 0, but the extrapolation gives $R(0) = 0.982$. The biggest source of deviation is probably the expected path length, computed from $\langle L \rangle = \frac{V}{A}$. The incoming beam profile is not uniform and has a higher intensity towards the center, as seen in Figure 4.1b. This means that the side walls and lids receive less incoming beam than the walls directly in front. The Born approximation therefore uses an overestimated average path length, giving a higher expected intensity, and thus a ratio less than one.

Parameter	Value
c	0.48323729
A	0.4993811
k [cm ⁻¹]	57.689165
p	0.57
$R(0) = c + A$	0.98261838
χ^2_ν	19.541338
Fit points	15
Fit parameters	4

Table 4.2: Stretched-exponential fit parameters for the observed/Born ratio.

4.3 Full Instrument Test with PSI Geometry

A high statistics simulation was performed with the PSI instrument file. It ran for more than 60 hours on 7 cores simulating $1 \cdot 10^{11}$ neutrons. The spherical monitor was corrected for latitude-dependent solid angle. To eliminate the numerical singularity close to the poles and the north-south asymmetry, the outermost 30° of each pole was discarded. This simulation showed an abnormal amount of asymmetry that cannot be explained, which is why 30° were removed. The heatmap is shown in Figure 4.2

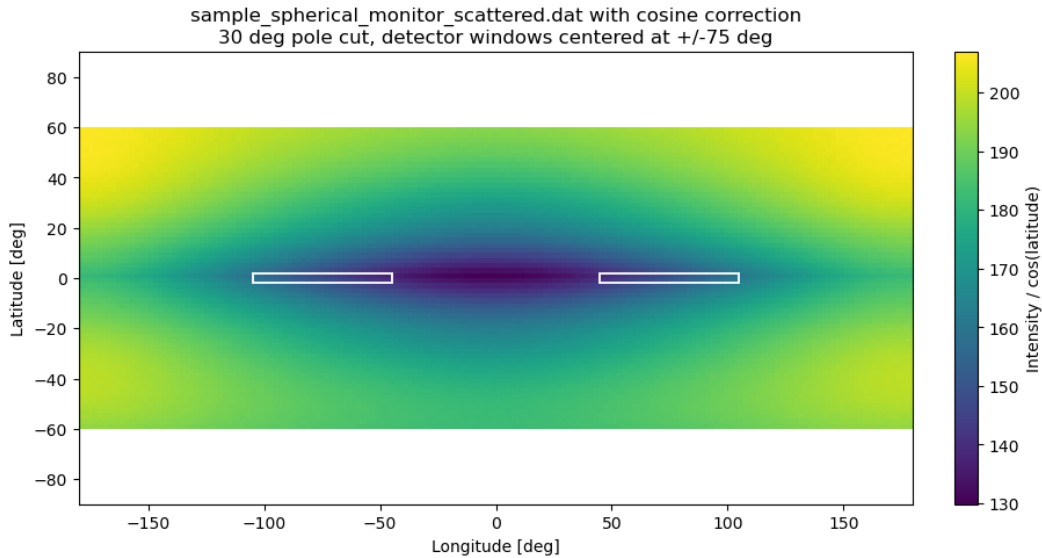
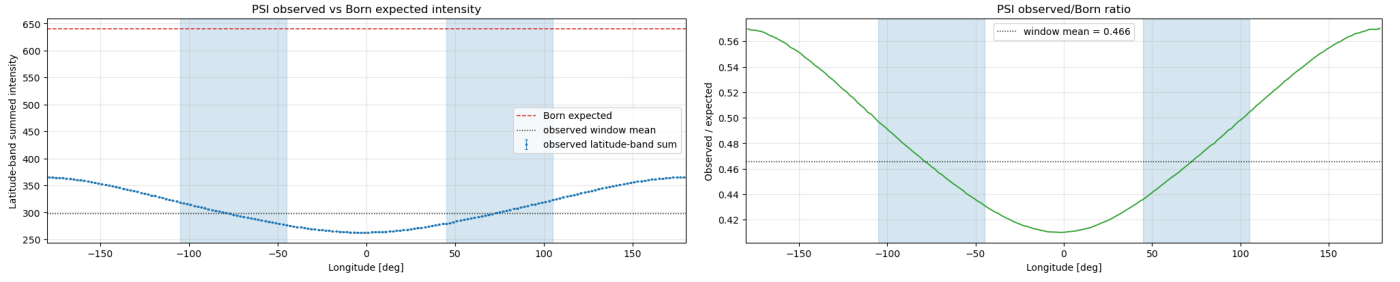


Figure 4.2: The heatmap of the PSI instrument with same vanadium geometry as PSI.

The heatmap shows a large intensity reduction in the forward beam direction. We extract an equatorial latitude band average to better understand what happens at the equator line. Comparing each bin average to the expected intensity computed from the Born estimate, we find the ratio 4.9 but plotted in 2ϕ



(a) Equatorial latitude band average compared to the Born (b) Angular latitude bin average ratio of observed to expected intensity.

Figure 4.3: Comparison of the angular intensity distribution with the Born approximation and the corresponding observed-to-expected ratio.

The observed **intensity with the Born** approximation is shown in Figure 4.3a and the ratio as a function of longitude is shown in Figure 4.3b. The Born estimated summed intensity within the two $60^\circ \times 4^\circ$ windows and the observed intensity were, respectively,

$$I_{\text{exp}} = 39731.5 \quad \text{and} \quad I_{\text{obs}} = 18509.0 \pm 0.9 \quad (4.13)$$

This gives an observed to expected ratio of 0.465853 or $\sim 46.6\%$. The reduction is because the Born approximation only accounts for single scattering events and assumes that all neutrons are scattered anisotropic. The ratio can be interpreted as an effective self-absorption factor for the selected angular window. We now evaluate the scattering produced using the quality metrics discussed in Section 4.2. The mean intensity within an equatorial latitude band and its corresponding standard deviation was

$$\bar{I} = 298.532 \pm 0.014 \quad \text{and} \quad \sigma_I = 13.12, \quad (4.14)$$

giving a coefficient of variation of

$$CV = 0.0439577. \quad (4.15)$$

The reduced chi-square value obtained by fitting a constant c to the two windows was the following

$$\chi_\nu^2 = 13317. \quad (4.16)$$

The coefficient of variation together with the reduced chi-squares shows that assuming isotropic spread within the detector arm is wrong. We can further quantify how wrong this is by looking at the peak-to-peak variation and the maximum absolute deviation from the mean. The peak-to-peak variation relative to the mean was

$$\frac{I_{\text{max}} - I_{\text{min}}}{\bar{I}} = 0.156791, \quad (4.17)$$

and the maximum absolute deviation from the mean was

$$\frac{\max |I - \bar{I}|}{\bar{I}} = 0.082544. \quad (4.18)$$

showing that the residual structure across the window is significantly larger than expected from counting uncertainty alone. The windows show clear anisotropy with a high reduced chi-square from a mean fit, and a peak-to-peak difference of $\sim 15.7\%$ and a maximum absolute deviation of $\sim 8.25\%$.

4.4 Optimization Study and Recommended Geometry for Calibration Use

4.4.1 Objective of the Optimization study

The objective of this chapter is to identify vanadium sample geometries that provide a strong calibration signal while remaining as close to an isotropic scattering distribution. This requires studying not only the scattered intensity but also the angular flatness of the scattering pattern and the deviation from the expected single-scattering intensity. The optimized geometry is identified with simulations using the Union framework in McStas.

4.4.2 Pareto optimization

Multi objective optimization or Pareto optimization is a branch of optimization that **perfectly** suits our goal. It is used when more than one objective that is to be minimized or maximized is involved, in our case, the flatness metric and the mean intensity. The answer is a set of possible solutions that give the best trade off between competing objectives [12, p. 2]. In a single objective search, the answer is easily determined by comparing the objectives to each other. For multi objective searches the goodness of a solution is instead determined by dominance. Dominance is defined such that solution x_1 dominates x_2 if x_1 is no worse than x_2 in all objectives or x_1 is strictly better than x_2 in at least one objective [12, p. 5]. The non-dominated solution set is the set of all solutions that are not dominated by any other solution. This non-dominated solution set is called the Pareto optimal solution set. The Boundary around the Pareto solution set forms the Pareto front [12, p. 7].

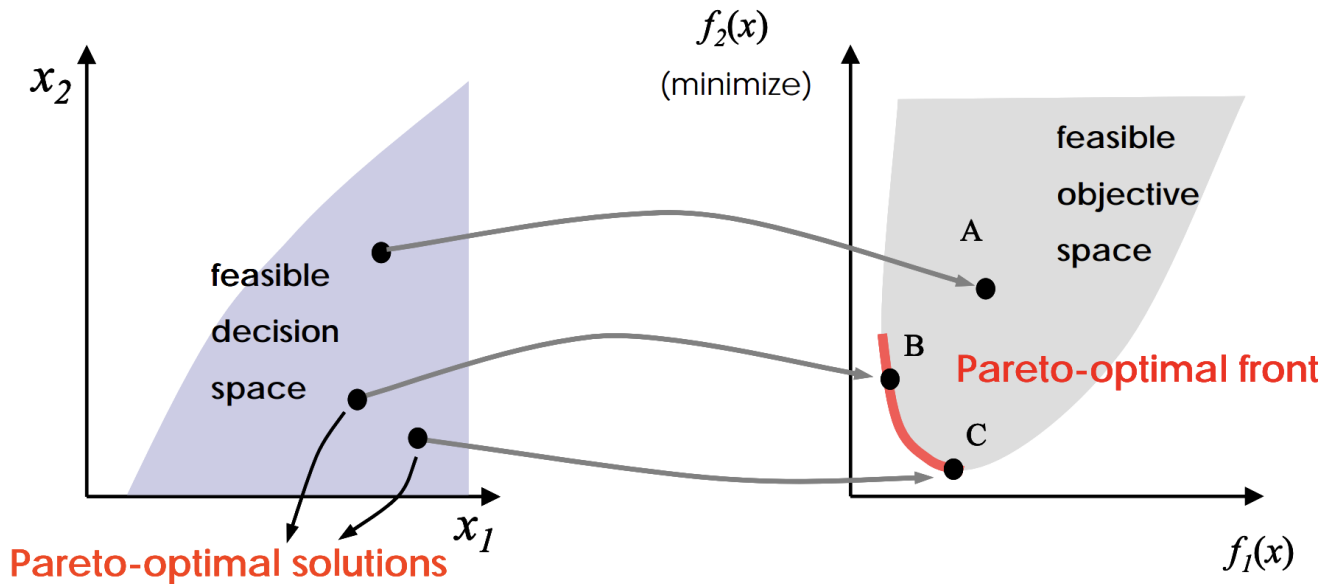


Figure 4.4: Illustration of the mapping from feasible solutions in the decision space to the objective space. Reproduced from [12, p. 8].

Filtering by practical geometry constraints. It makes sense to look for unique solution sets, since the Pareto solution set can give solutions that are alike. By unique we mean that if a sample with the same radius and height but multiple different thicknesses or window offsets is found in the Pareto optimal solution set, then we only choose one of them.

4.4.3 Recommended Geometry for Calibration Use

Candidate geometries. Performing simulations in a wide range of height, radius, and thickness should provide a good estimate for the best geometry. The geometry ranges were $H = (10, 60)$ [mm], $R = (5, 15)$ [mm]

and $T = (1.0, 4.0)$ [mm]. Computing the reduced chi-square and the total intensity within the two windows, and performing multi objective optimization, we find the non-dominated solution set, shown in Figure 4.5

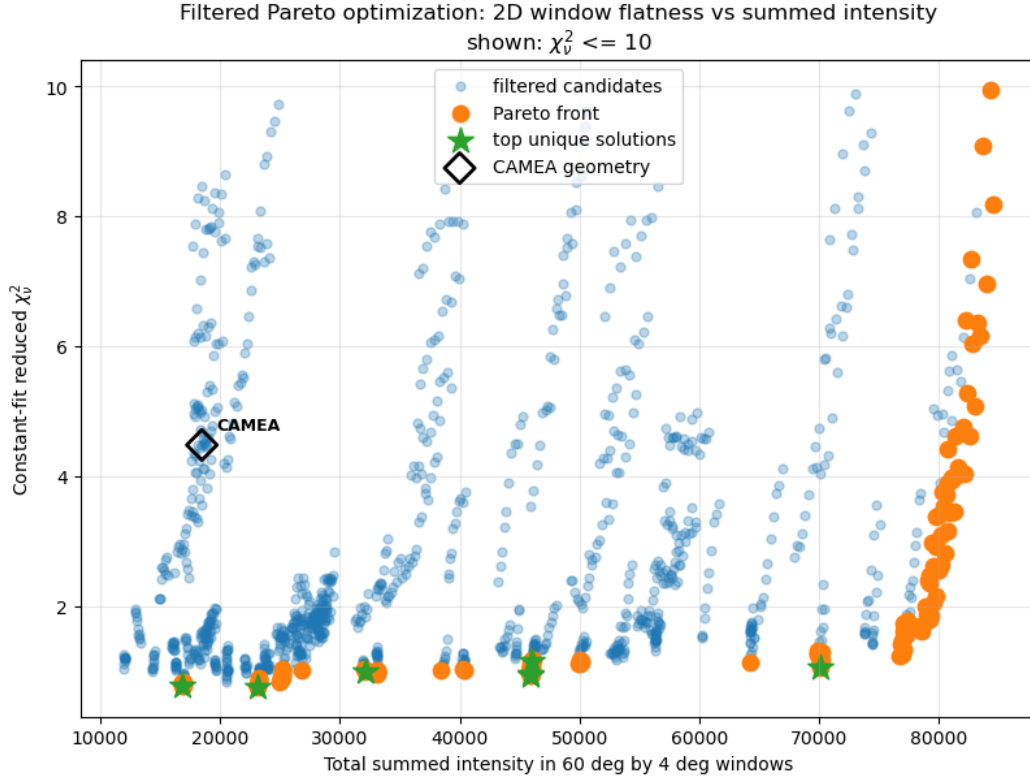


Figure 4.5: Solution set for various geometries.

In total, we scan 181 different geometries with a movable detector arm, giving a solution set of 1793 candidates. Computing the non-dominated solution set, we find a Pareto solution set with a size of 148 candidates. Of that, 7 solutions are unique in terms of thickness and offset. They are shown in Table 4.3

Rank	r [mm]	h [mm]	t [mm]	λ [Å]	Offset [deg]	χ^2_ν	Sum
1	5.00	50.00	1.75	4.04	−20.0	0.76002	2.309×10^4
2	5.00	40.00	1.00	4.04	−4.0	0.78430	1.688×10^4
3	10.00	40.00	1.75	4.04	−20.0	0.93155	4.586×10^4
4	10.00	50.00	1.75	4.04	−8.0	0.94714	4.592×10^4
5	10.00	30.00	1.00	4.04	+20.0	1.00620	3.210×10^4
6	15.00	50.00	2.12	4.04	−8.0	1.06260	7.007×10^4
7	15.00	40.00	1.00	4.04	+0.0	1.15550	4.605×10^4

Table 4.3: Top unique Pareto solutions after filtering.

We observe that some of the geometries give $\chi^2_\nu < 1$. In general, this can suggest that the uncertainties are too large. However, reduced chi-squares near unity are not surprising in this study because the uncertainties are estimated from finite Monte Carlo simulations. Only $5 \cdot 10^7$ neutron rays are used to estimate the intensity for each geometry. The resulting statistical estimates of intensity and uncertainty are only approximate. The reduced chi-square should therefore be used more as a relative measure of flatness than as an absolute goodness-of-fit criterion. The cosine corrected intensity heat map for rank 1 is shown in Figure 4.6a and rank 3 is shown

in Figure 4.6b. Rank 1 and rank 3 exhibit an almost complete isotropic spread within the 60° window and have been selected for their relatively high intensity.

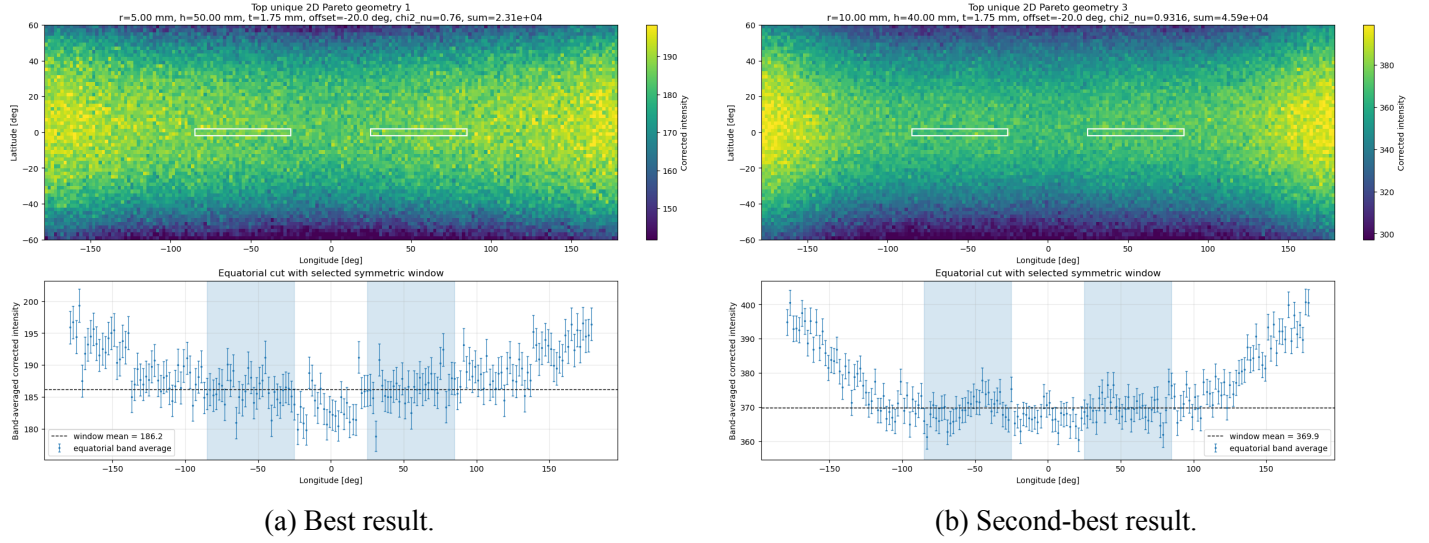


Figure 4.6: Comparison of the two best optimization results.

Deviation From the Expected Single-Scattering Intensity. Using the same equator cut as in section 4.3, and computing the expected intensity per bin, we find the ratio in Figure 4.7

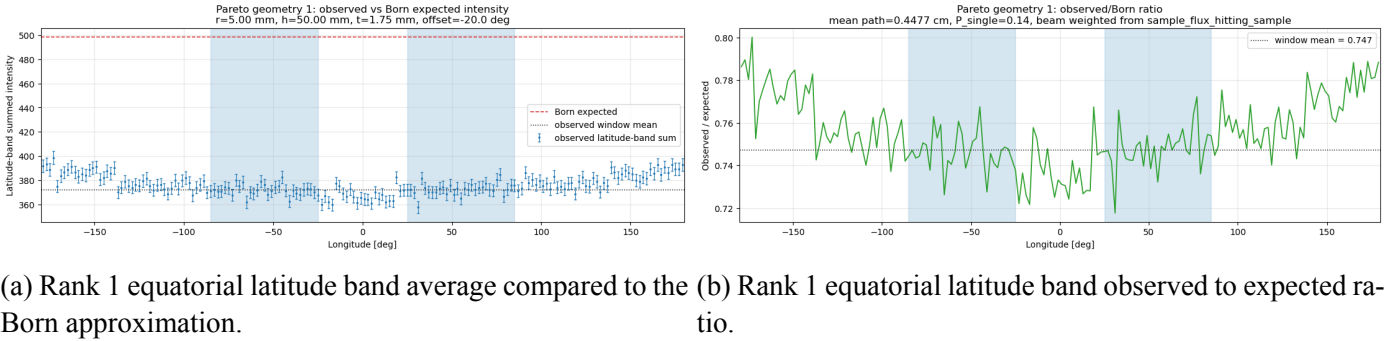


Figure 4.7: Comparisons of the angular intensity distribution of the rank 1 geometry to the Born expected intensity distribution.

For the optimized geometry, we see an almost completely flat distribution and less deviation from the born approximation. A generally higher ratio compared to the CAMEA-sample is observed, indicating that less neutron rays are scattered in different directions or absorbed. Overall, this shows a more effective sample geometry because it scatters close to isotopically within the region and has lower self-absorption.

5 Discussion

5.1 Physical interpretation of trends

Even though Vanadium is characterized by a dominating incoherent scattering cross section $\sigma = 5.08\text{barn}$ that comes from the fluctuations in scattering lengths, the resulting isotropy is in practice limited by the mean path length. This means that even if the individual scattering cross sections are isotropic, phenomena such

as self absorption and multiple scattering introduces angular dependence in the observed spread. It is therefore necessary to correctly model the anisotropy produced by samples of various geometries.

5.2 Limitations of Monte Carlo model

Where McMini runs as a pure Monte Carlo physics simulation, McStas weighs neutron ray absorption, rather than removing absorbed rays, as a variance reduction technique. This contrast is the biggest difference between the two engines. McMini works as a minimalist validation tool that removes absorbed neutron rays. This results in larger statistical noise, which will affect the reliability of the equator cut and meridian cut. However, since the difference between the two ray tracing software can be attributed to noise, this is a strong result because it shows that the variance reduction implemented in McStas works.

5.3 Statistical precision

The observed integrated intensity across the two ray tracing programs agree down to the 10^{-4} . This level of agreement proves that the Union framework's introduction of Monte Carlo weight and ray absorption weights does not introduce a physical bias. This level of trust to the framework is crucial when simulating and isolating physical properties of a system.

5.4 Limits of Union

The main limitation observed in McStas was the north-south asymmetry in the PSD_monitor_4PI component. The asymmetry levels for the cylindrical vanadium sample were $1.49 \cdot 10^{-2}$ and $3.6.2$ and for the spherical vanadium sample it was $1.43 \cdot 10^{-2}$. This shows that the asymmetry was consistent across geometries. Furthermore, the analysis into the asymmetry with only a downstream beam onto a flat monitor and a spherical monitor, showed that the asymmetry was only present for the spherical monitor. This points towards the problem stemming from coordinates binning or weight redistribution. The consequence of this redistribution error is that it compromises the detector efficiency near the poles. To minimize this error it is necessary to apply a $5^\circ - 10^\circ$ pole cut.

6 Conclusion

6.1 Validation summary

The validation of the Union implementation in McStas shows that correct physical behavior was reproduced. The total integrated intensity showed a bin-per-bin residual on the order of $1.86 \cdot 10^{-11}$. The Union framework correctly decomposes neutron scattering of order 1 up to 5. The asymmetry diagnostics,

6.2 Suggested Geometries and Pareto Optimization Results

The problem that the Pareto optimization solves is how isotropic can a 60° window get while still producing enough intensity spread. The flatness is highly correlated with the sample volume and wall thickness, however, small samples are undesirable as their cross sections are small and thus produce little intensity. The multi objective optimization search finds the non-dominated solution set in a search of over 400 geometries that display flatness and high mean intensity within the detector window. Based on the search the top 2 ranked geometries were,

- Rank 1: $r = 5[\text{mm}]$, $h = 50[\text{mm}]$, $t = 1.28[\text{mm}]$, $\text{offset} = 0^\circ$
- Rank 2: $r = 10[\text{mm}]$, $h = 60[\text{mm}]$, $t = 1.87[\text{mm}]$, $\text{offset} = -10^\circ$

In general, the deviation for the unique Pareto solution set from the Born approximation was $\sim 30\%$. The thin limit gave an "observed/expected" ratio of 1.06. The deviation from unity comes for the path length estimation, $\langle L \rangle = V/A$. By choosing geometries with thinner walls the deviation from the Born approximation decreases. With a ratio of $\sim 70\%$ for the unique Pareto solution set and $\sim 46.8\%$ for the vanadium sample simulated with the PSI instrument, we see a significant improvement in efficiency.

6.3 Future work

Correcting the asymmetry artifact and producing a spherical monitor that correctly weighs and bins data is critical. Any simulation data recorded with the component `PSD_monitor_4PI` should discard at least 10° of the data away from the poles. In version 3.6.14 in `McStas` this problem persists. The component `Cyl_monitor_PSD` has a -1 angular bin shift and subsequently records 0 intensity in the highest angular bin. The monitor data should be shifted by 1 bin to the right and the lowest angular bin is to be removed since it now has 0 intensity however, the data will be left to right symmetric.

Investigating more complex Union environments and Union geometries. Experimentally verifying that the number 1 ranked geometry produces the simulated data.

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A Theoretical background derivations

A.1 Derivation of Path Length Sampling

Divide the sample into thin slices of area A and thicknesses dz , such that the total scattering cross section is much smaller than the area perpendicular to the beam. Essentially so small that a particle cannot obscure another particle within the same slab. This is what is known as the Born approximation. The total cross section of a slice is then,

$$\Sigma_t = nA\sigma dz \quad (\text{A.1})$$

Divide by the beam area normalized such that beam area is equal to the area of the slab.

$$\frac{\Sigma_t}{A} = n\sigma dz \quad (\text{A.2})$$

That was the probability of interaction for a slab. Now let $p_t(z)$ be probability of transmission through the slab for a neutron. Then the probability that a neutron will be transmitted distance dz is $(1 - n\sigma dz)$. The probability of traversing dz without interaction is

$$p_t(z + dz) = p_t(z)(1 - n\sigma dz) \quad (\text{A.3})$$

Expand and move terms around to find

$$\frac{p_t(z + dz) - p_t(z)}{dz} = -p_t(z)n\sigma \quad (\text{A.4})$$

Where we recognize the LHS as the definition of a derivative.

$$\frac{dp_t(z)}{dz} = -p_t(z)n\sigma \quad (\text{A.5})$$

This is a simple first order differential equation. Solving we find the probability of a neutron being transmitted a distance z ,

$$\int \frac{1}{p_t(z)} = -n\sigma \int_0^z dz' \quad (\text{A.6})$$

Isolating for the probability

$$\ln(p_t(z)) = -n\sigma z \quad (\text{A.7})$$

$$p_t(z) = \exp(-n\sigma z) = \exp(-\Sigma_t z) \quad (\text{A.8})$$

This is the expression that we are familiar with from the neutron scattering notes. The probability of an interaction is then simply

$$p_{int}(z) = 1 - \exp(-\Sigma_t z) \quad (\text{A.9})$$

We then sample a random number uniformly in $U[0, 1]$

$$p_{int}(z) = 1 - \exp(-\Sigma_t z) = U[0, 1] \quad (\text{A.10})$$

This gives,

$$z = -\frac{\ln(1 - U)}{\Sigma_t} \quad (\text{A.11})$$

Since U is uniformly distributed, then $1 - U$ is also uniformly distributed about a reflection around 0.5. Due to this fact we can simplify our equation to

$$z = -\frac{\ln U}{\Sigma_t} \quad (\text{A.12})$$

Sampling from 0 to 1 will now allow us to randomly sample path lengths following the physical properties of the material.

A.2 Derivation of the expected per bin intensity

A value that is of great importance is the self-absorption. It comes from naively assuming only isotropic scattering and ignoring other effects such as multiple scattering and absorption. Start from the differential scattering cross section [1, p.13]

$$\frac{d\sigma}{d\Omega} = \frac{1 \text{ \# Neutron scattered into solid angle } d\Omega \text{ per sec.}}{\Psi} \quad (\text{A.13})$$

This is given per atom. Now we multiply by N_{nuc} to scale up to the full sample. We also convert the number of scattered neutron to flux-like units same units as flux used in McStas: $1/(s \cdot \text{cm}^2 \cdot \text{sr} \cdot \text{energy unit})$,

$$N_{nuc} \frac{d\sigma}{d\Omega} = \frac{1}{\Psi} \frac{dI}{dt d\Omega} \quad (\text{A.14})$$

For vanadium: $\frac{d\sigma}{d\Omega} = \frac{\sigma_{inc}}{4\pi}$,

$$N_{nuc} \frac{\sigma_{inc}}{4\pi} = \frac{1}{\Psi} \frac{dI}{dt d\Omega} \quad (\text{A.15})$$

Rearranging the equation above,

$$N_{nuc} \frac{\sigma_{inc}}{4\pi} \Psi dt d\Omega = dI \quad (\text{A.16})$$

where the number of nuclei is the number density times sampled volume $N_{nuc} = nV$

$$nV \frac{\sigma_{inc}}{4\pi} \Psi dt d\Omega = dI \quad (\text{A.17})$$

flux is given in per sec per square centimeters. So when integrating up we simply get a factor of 1 from integrating the flux. Integrating the solid angle is slightly trickier. We get

$$\int_{\theta_1}^{\theta_2} \int_{-\phi_0}^{\phi_0} \cos \phi d\phi d\theta = 2 \sin \phi (\theta_2 - \theta_1) \quad (\text{A.18})$$

which for bin spanning 2 degrees by 2 degrees is $1.218 \cdot 10^{-3} \text{sr}$. Since flux is defined as total intensity striking the sample I_{hit} over area A we insert that and get $\frac{V}{A}$ which is simply our expected path length through sample $\langle L \rangle$. We know that number density times microscopic scattering cross section is equal to macroscopic cross section. Number density in units of $[\text{cm}^{-3}]$ and microscopic cross section in units of $[\text{cm}^2]$. Together they make the macroscopic cross section Σ_{inc} in units of $[\text{cm}^{-1}]$ Insert this into Eq. A.17 we get

$$I_{bin} = I_{hit} \Sigma_{inc} \langle L \rangle \frac{\Delta\Omega}{4\pi} \quad (\text{A.19})$$

Where we recognize that $\Sigma_{inc} \langle L \rangle$ is simply linear term for interaction probability. To get a more precise estimate we instead can use $P(\langle L \rangle) \simeq 1 - \exp(-\Sigma_{inc} \langle L \rangle)$ Thus we expect intensity in bin spanning 2 degrees by 2 degrees

to be

$$I_{\text{bin}} = I_{\text{hit}} (1 - \exp(-\Sigma_{\text{inc}} \langle L \rangle)) \frac{\Delta\Omega}{4\pi} \quad (\text{A.20})$$

The equation uses macroscopic scattering cross sections, but we can define it with more precision by accounting for all neutron interactions and then taking the fraction of neutron interactions that scatter: $\Sigma_{\text{inc}}/\Sigma_{\text{t}}$. Then we obtain

$$I_{\text{bin}} = I_{\text{hit}} \frac{\Sigma_{\text{inc}}}{\Sigma_{\text{t}}} (1 - \exp(-\Sigma_{\text{t}} \langle L \rangle)) \frac{\Delta\Omega}{4\pi} \quad (\text{A.21})$$

Then define the probability factor as

$$P_{\text{scatt}} = \frac{\Sigma_{\text{inc}}}{\Sigma_{\text{t}}} (1 - \exp(-\Sigma_{\text{t}} \langle L \rangle)) \quad (\text{A.22})$$

We get the simple expression that tells us the expected intensity in a bin

$$I_{\text{bin}} = I_{\text{hit}} P_{\text{scatt}} \frac{\Delta\Omega}{4\pi} \quad (\text{A.23})$$

B Detailed Technical Derivations and Geometric Proofs for McMini

The detailed technical derivations adds some key details to better convey exactly how the monitors compute intersections and bins the data. There is not enough space in the main thesis to go into detail, but for the sake of transparency, the exact methods are outlined below.

B.1 Derivation of flat monitor intersection and binning

The first monitor that was implemented in McMini was a flat position sensitive monitor. It is the simplest monitor to add, and it serves as a source validation at a very base level. Base level is to be understood as a spatial and wavelength distribution level.

Ray-plane intersection

The flat plane monitor is a flat rectangular surface, placed perpendicular to the beam direction. For each neutron history the intersection between the ray trajectory and the flat plane was computed and binned. The ray trajectory is parametrized by

$$\vec{r}(t) = r_0 + \vec{v} \cdot t, \quad (\text{B.1})$$

where $\vec{r}_0$ is the initial position vector and $\vec{v}$ is the direction vector. For simplicity we will not allow the monitor to be rotated in any way. This means we place the monitor at some fixed position, which we will call $z = z_{\text{mon}}$

$$z_0 + t v_z = z_{\text{mon}}, \quad (\text{B.2})$$

which gives

$$t = \frac{z_{\text{mon}} - z_0}{v_z}. \quad (\text{B.3})$$

Here, some sorting is done to remove all t -values that are of no interest. That would be when $t < 0$ or if $v_z \leq 0$. The first is a safeguard to only consider forward intersections. The other condition removes rays that are parallel or facing away from the monitor plane. Once sorting is done the hit position is computed using the ray trajectory. This point is used for binning.

Spatial binning

The monitor surface is a 2-dimensional grid of $n_x \times n_y$ pixels covering the region

$$x \in [-\text{xwidth}/2, \text{xwidth}/2], \quad y \in [-\text{yheight}/2, \text{yheight}/2]. \quad (\text{B.4})$$

The hit position $(x_{\text{hit}}, y_{\text{hit}})$ is assigned to pixel indices by first normalizing to the unit interval:

$$f_x = \frac{x_{\text{hit}} - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}, \quad f_y = \frac{y_{\text{hit}} - y_{\text{min}}}{y_{\text{max}} - y_{\text{min}}}. \quad (\text{B.5})$$

This gives us a normalized hit position in the unit interval that can be given a pixel value and thus a place on the heatmap

$$i = \lfloor f_x n_x \rfloor, \quad j = \lfloor f_y n_y \rfloor. \quad (\text{B.6})$$

Hits falling outside this range are simply discarded. For numerical stability we also clamp the indices to valid range of $[0, n_x - 1]$ and $[0, n_y - 1]$. Each valid hit can now contribute its neutron weight to the appropriate pixel. This forms a two-dimensional histogram of intensity.

C Supplementary Validation Data: Solid Vanadium Sphere

C.1 Spatial intensity comparison

Raw intensity maps The first part of the validation will be a comparison between the spatial intensities of McMini and McStas. The simplest geometry to implement in C was a sphere, so the first section will be on a solid sphere. All parameters used in McStas are shown below and McMini settings were set to match. Parameters not mentioned can be assumed to be the default values of McStas [9] The raw monitor data from

Component	Parameters
Source_simple	$\lambda_0 = 4.0, \Delta\lambda = 0.1, \text{focus_xw} = 0.02, \text{focus_yh} = 0.02, \text{dist} = 1.0, \text{radius} = 0.01, n = 10^9$
Incoherent_process	$\sigma = 5.08, \text{unit_cell_volume} = 13.827$
Union_make_material and Union_sphere	$\text{my_absorption} = 36.7, \text{radius} = 0.01$
PSD_monitor_4PI	$\text{nx} = 360, \text{ny} = 180, \text{radius} = 1.0, \text{restore_neutron} = 1$

Table C.1: Simulation parameters for the spherical vanadium sample.

McMini and McStas is seen in Fig C.1. At first look they are indistinguishable. Both exhibit the expected angular dependence and a decrease of intensity towards the poles.

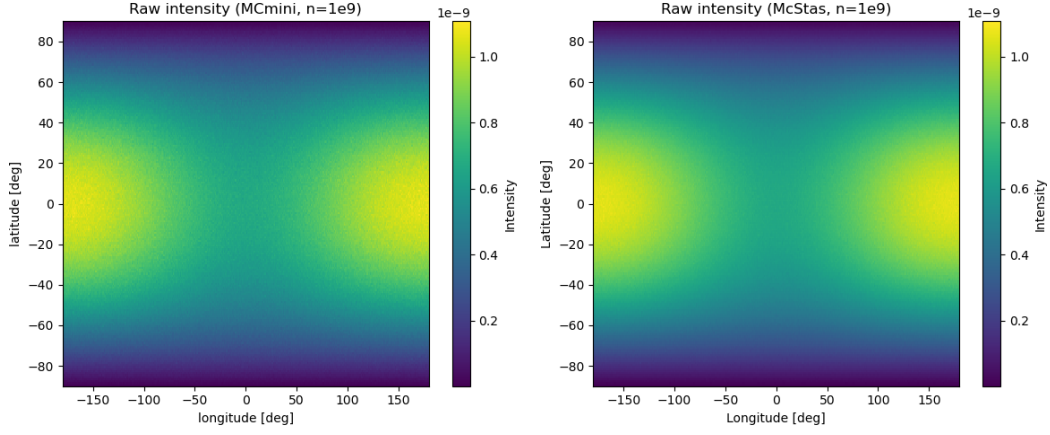


Figure C.1: Raw intensity map of a solid sphere with radius of 1cm

The total intensities obtained from both simulations are

$$I_{\text{McMini}} = 3.6189 \times 10^{-5}, \quad I_{\text{McStas}} = 3.6195 \times 10^{-5}.$$

The ratio is

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} = 0.99984,$$

demonstrating excellent normalization agreement at the 10^{-4} level.

Difference map To correct for solid angle, each bin was divided by the geometrical $\cos(\text{latitude})$ factor. The corrected maps are shown in Fig. C.2. The poles include a higher degree of noise, since the bins are smaller and thus receive fewer rays. Once the correction is applied this noise can blow up. Therefore the data within 5 degrees of the poles has been removed. Then subtracting the two maps gives the difference map in Figure C.2

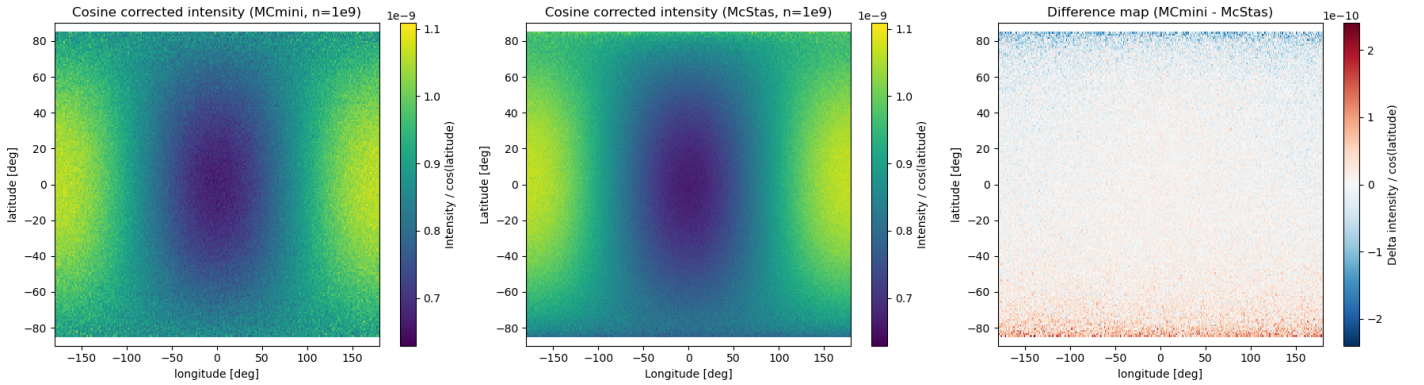


Figure C.2: Corrected intensity map and difference map of a solid sphere with a radius of 1cm

The difference map shows mostly noise, but some clear bias is visible. A north bias, which is believed to come from the 4PI monitor, shows itself. This is where the asymmetry was initially discovered. Quantifying to difference map, using mean residuals, RMS residuals, mean pull, and reduced chi-square gives the values in Table 3.2

Metric	Cosine-corrected map
Mean residual	$-1.562070 \times 10^{-13}$
Mean absolute residual	2.216481×10^{-11}
RMS residual	3.149105×10^{-11}
Mean pull	-5.172632×10^{-3}
Pull standard deviation	1.142612
Reduced χ^2	1.305589

Table C.2: Quantitative comparison of the cosine-corrected difference map between McMini and McStas.

For the cosine corrected map the mean residual ($-1.56 \cdot 10^{-13}$) shows agreement in total intensity, while the residual RMS (3.15×10^{-11}) shows a typical per bin mismatch on the order of 10^{-11} . The pull width was 1.14 and the reduced χ^2 was 1.31, showing some excess over pure statistical noise. This excess being the asymmetry that is visible, and not enough to say that the simulations do not match.

Equator and meridian cut Taking a cut along the equator and meridian shows the agreement along a line of bins. The result is seen in Figure C.3

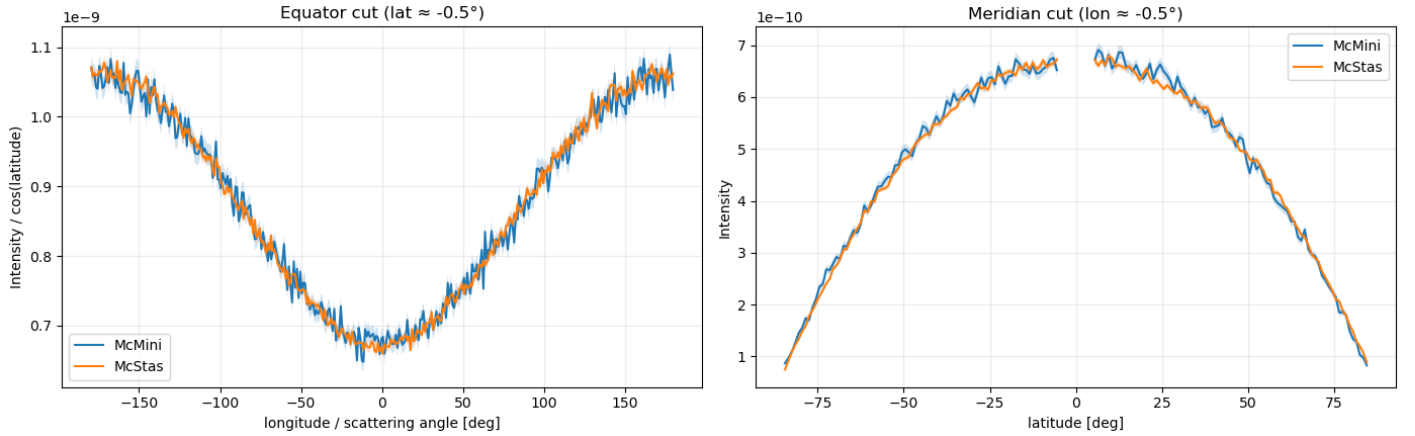


Figure C.3: Equator and meridian cut for a solid sphere with a radius of 1cm

The cuts show excellent agreement for both the equator and meridian cut. McMini displays greater levels of noise, but this is because it does not use the same variance reduction method that McStas uses.

Integrated intensities After applying the 5° pole cut and the cosine correction, the total integrated intensities of the full 4π maps are

$$I_{\text{McMini}} = 5.372803 \times 10^{-5}, \quad I_{\text{McStas}} = 5.373468 \times 10^{-5}.$$

The ratio is

$$\frac{I_{\text{McMini}}}{I_{\text{McStas}}} = 0.9998762,$$

Integrating the northern and southern hemisphere.

$$N_{\text{McMini}} = 2.686408 \times 10^{-5}, \quad S_{\text{McMini}} = 2.686395 \times 10^{-5},$$

and

$$N_{\text{McStas}} = 2.725139 \times 10^{-5}, \quad S_{\text{McStas}} = 2.648329 \times 10^{-5}.$$

This corresponds to hemisphere asymmetries

$$A_{\text{McMini}} = \frac{N - S}{N + S} = 2.484464 \times 10^{-6}, \quad A_{\text{McStas}} = \frac{N - S}{N + S} = 1.429436 \times 10^{-2}.$$

showing a clear asymmetry for the hemisphere, while the total intensity agree excellently.

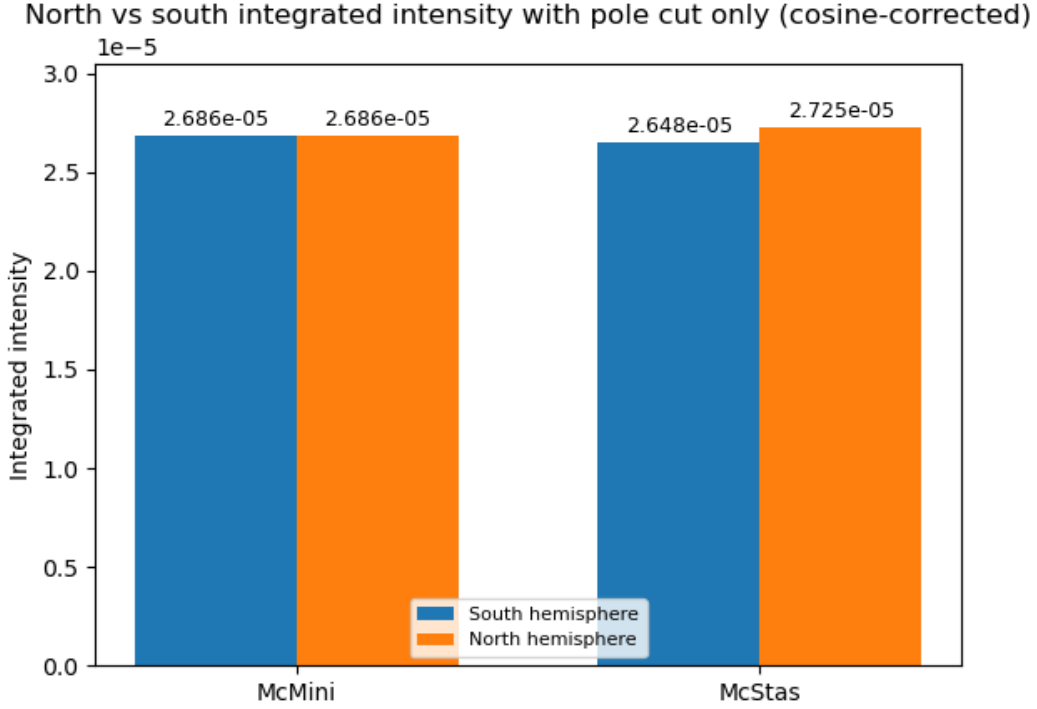


Figure C.4: Cosine-corrected integrated intensities in the northern and southern hemispheres

A clear asymmetry is present in the McStas result, whereas the McMini result remains essentially symmetric. As was discussed in Section D.2.2, asymmetry is present in the 4PI monitor. The largest part of the asymmetry observed here, seems to come from that issue.

C.2 Scatter-order analysis

Last part of the validation for the solid sphere will be on intensity maps of neutrons with different scatter orders. McMini has been made with the ability to count interactions just like McStas. Figure C.5 shows the intensity maps of neutrons with scatter order equal to 1, 2, 3, 4, and ,5.

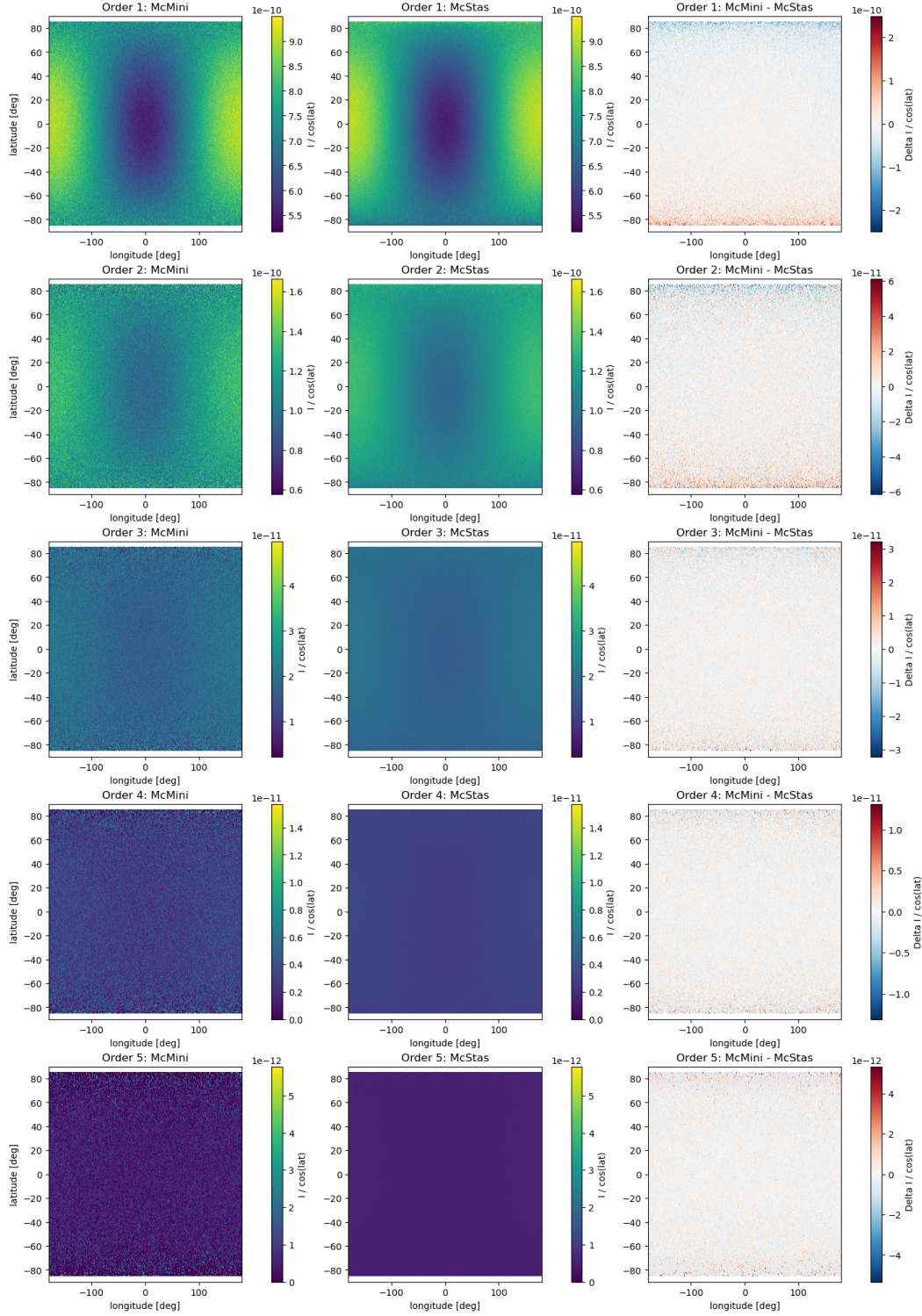


Figure C.5: Scatter order maps for McStas and McMini, together with their difference map

Visually, all scatter orders match great. For high orders, such as 5 the higher noise in McMini due to no variance reduction becomes very clear. Table C.3 shows integrated intensity and intensity ratios of different scatter orders. It also shows the percentage share that a scatter order makes up of the total intensity. For all orders, integrated intensity are in unity. They agree on the share of intensity, and RMS shows low per-bin disagreement. The asymmetry is still present, but seems to be less for higher orders.

Table C.3: Cosine-corrected, order-resolved integrated intensities for the sphere case with pole cut $|\text{latitude}| \leq 85^\circ$. Shares are given relative to the summed intensity over orders 1–5.

Order	I_{McMini}	$\text{Share}_{\text{McMini}}$	I_{McStas}	$\text{Share}_{\text{McStas}}$	$I_{\text{McMini}}/I_{\text{McStas}}$	RMS
1	4.536851×10^{-5}	84.45%	4.537335×10^{-5}	84.45%	0.9998934	2.824796×10^{-11}
2	6.990619×10^{-6}	13.01%	6.991936×10^{-6}	13.01%	0.9998117	8.611974×10^{-12}
3	1.138327×10^{-6}	2.12%	1.139436×10^{-6}	2.12%	0.9990266	3.215086×10^{-12}
4	1.912518×10^{-7}	0.36%	1.908454×10^{-7}	0.36%	1.0021300	1.297401×10^{-12}
5	3.250116×10^{-8}	0.06%	3.241492×10^{-8}	0.06%	1.0026600	5.295745×10^{-13}

D Technical Note on the 4π Monitor Asymmetry

D.1 Structural analysis on the cause of the observed asymmetry

This asymmetry artifact was initially discovered during the multiple scattering simulation with spherical and cylindrical samples. This caused some confusion since it was unexpected that symmetrical sampled would yield asymmetrical results. This problem was investigated and has been narrowed down to a PSD_monitor_4PI issue. A brief mention of other factors that were investigated since they could have caused the issue included:

1. **Incorrect monitor selection / inclusion of unscattered rays, and beam-stop treatment.** The possibility that the asymmetry came from the transmitted neutron, or from incorrect beam-stop handling was tested by restricting the analysis to only scattered neutrons, i.e. using $n_{\text{reflections}} > 0$ in McStas and McMini. The asymmetry persisted so was not caused by this.
2. **Cosine correction of the 4π monitor.** The monitor maps were compared before and after correction for the solid angle. The asymmetry was already present in the raw data, so the effect is not caused by the cosine correction.
3. **Monitor geometry in McMini.** The McMini spherical monitor was tested in two modes. geometric sphere-hit binning and direction-only binning. This changed some details, but the discrepancy could not be explained as a geometric artifact on the McMini side.
4. **Incorrect uncertainty model.** The McMini monitor was extended to record weighted uncertainties using $\sqrt{\sum w_i^2}$, instead of a Poisson assumption. This improved statistics but did not reduce asymmetry.
5. **Source asymmetry on a flat monitor** As was already shown, no asymmetry showed itself for a flat monitor. This includes a flat monitor immediately after the source and downstream. With a north to south asymmetry at the level of 10^{-6} to 10^{-5} and a negligible fitted vertical slope
6. **Absorption / wavelength-dependent attenuation.** The scattering experiment was repeated in McStas with absorption set to 0. The asymmetry remained.
7. **Scatter-order dependence.** The asymmetry remained for one, two, and three scattering events. The asymmetry was strongest for the single scattering event and was reduced to mostly noise for higher order.
8. **Simple indexing or sign-convention errors.** Tests involving latitude flips, longitude flips, and ± 1 -bin shifts did not remove the residual structure.
9. **Bare 4π monitor response without a sample.** This is what was shown in Figure D.1, and seems to be the main cause of asymmetry. This configuration agrees in total intensity but favors a north to south bias.

Thus, the asymmetry is not caused by inclusion of unscattered rays, beam stop handling, cosine correction, source non uniformity, absorption, or a simple indexing mistake. The evidence points to an issue in the McStas 4π monitor, which favors a north bias.

D.2 Asymmetry analysis on the flat monitor vs. spherical monitor

D.2.1 Flat monitor

Comparison with McStas flat monitor

To verify the flat monitor implemented in McMini, a comparison was made with an instrument in McStas that contained only a source and a flat PSD_monitor was made. Instrument parameters used in McStas and matched in McMini were:

Component	Parameters
Source_simple	$\lambda_0 = 4.0$, $\Delta\lambda = 0.1$, focus_xw=focus_yh= 0.5, dist=1.0, radius= 0.25, $n = 10^9$
PSD_monitor	xwidth=yheight= 1.0, nx=ny= 2000, restore_neutron= 1

Table D.1: Simulation parameters for the source and planar PSD monitor.

The comparison will include visual inspection, their difference map, and a check for asymmetry. Since the sources are equal we expect only noise to be left when subtracting the two, and since it is symmetric, we would expect only noise when mirroring the south onto the north side. Raw data and the difference map can be seen in Figure D.1.

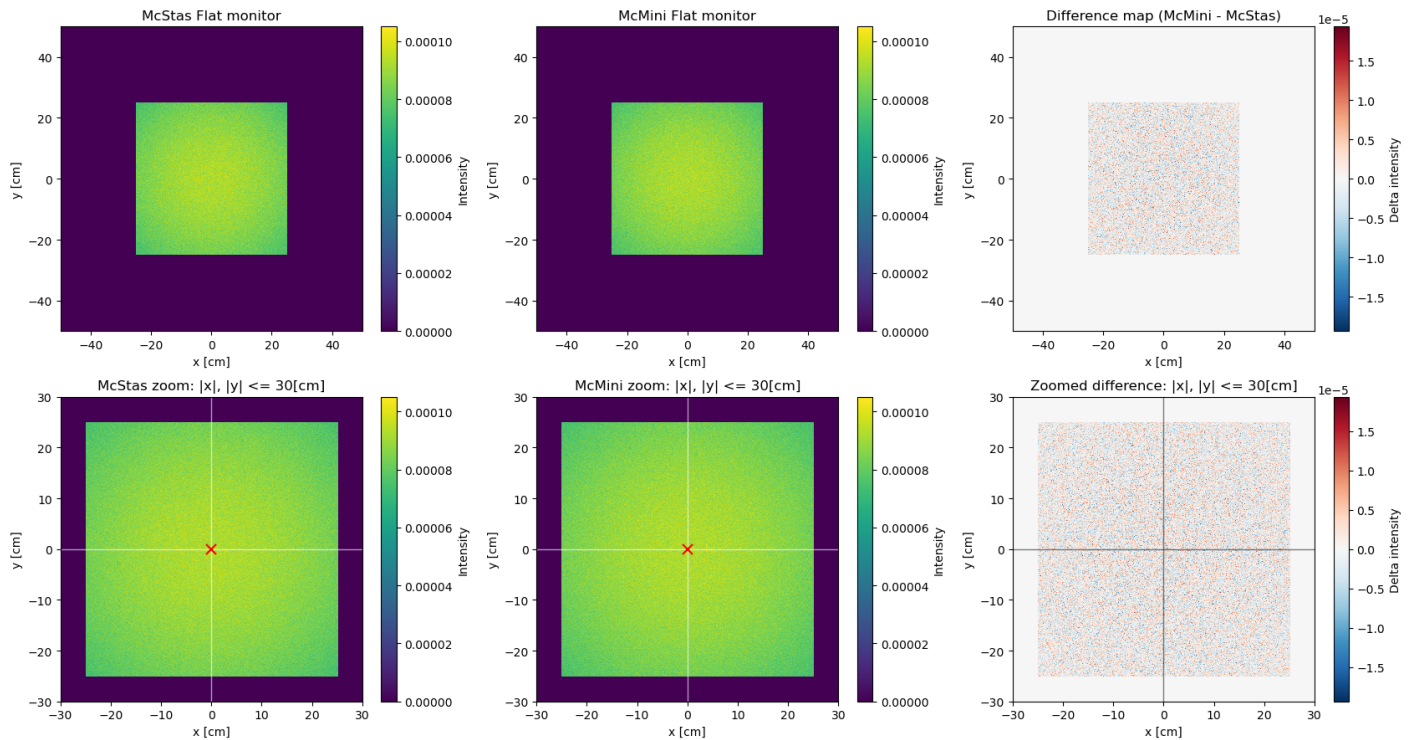


Figure D.1: Comparison and zoomed comparison of McStas and McMini

The same color scale has been forced as is standard practice. Visually they are indistinguishable, and then subtracting the two leaves nothing but noise on the difference map.

Flat monitor symmetry diagnostics

Since the geometry is symmetric, no north to south or east to west asymmetry is expected. To verify that no artificial directional bias is introduced at the source or detection level, symmetry test were performed by mirroring the data onto itself.

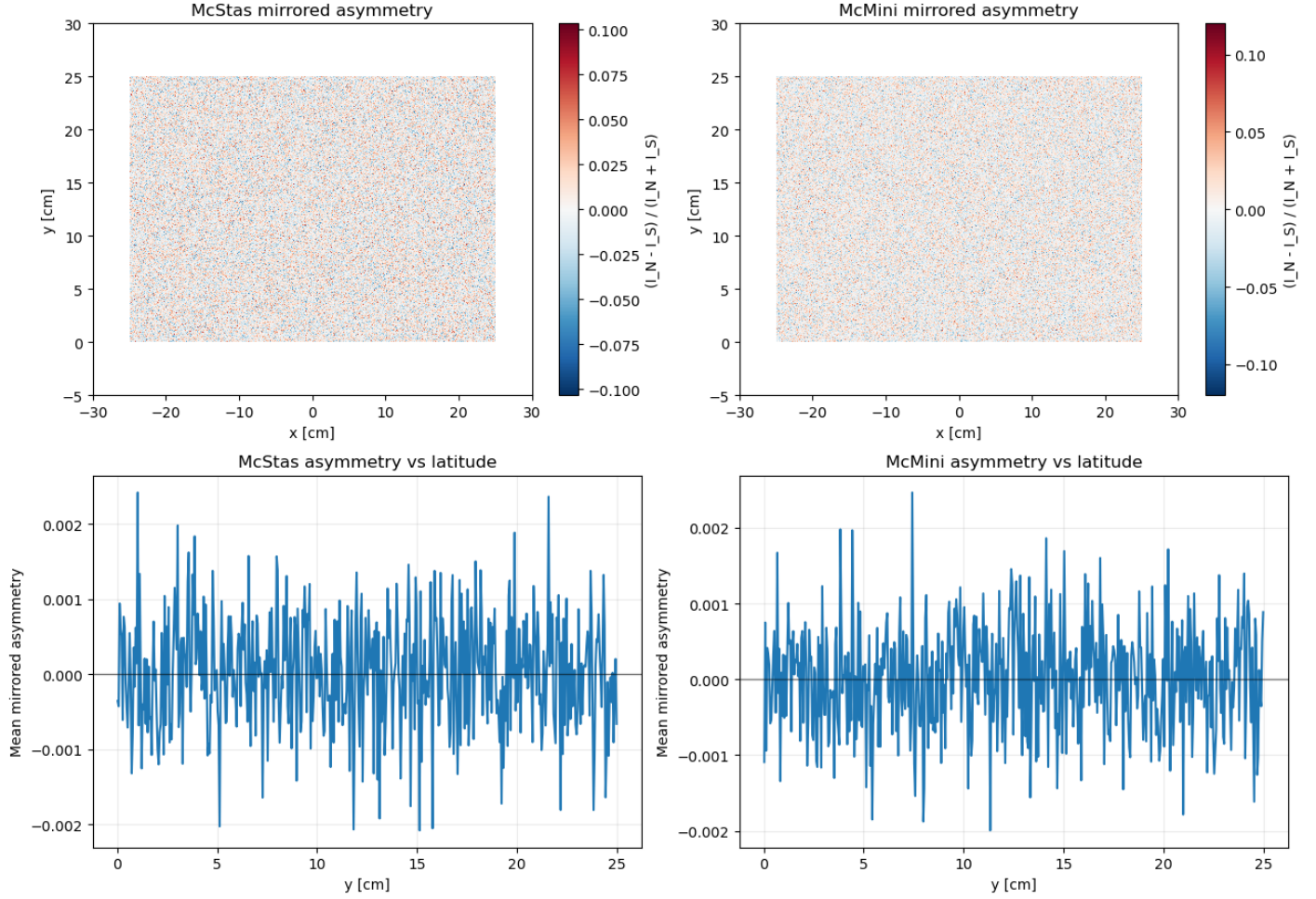


Figure D.2: Asymmetry analysis of McStas and McMini

Mirroring the data onto itself shows no signs of asymmetry, and both mean data along each axis show only noise around 0.

Integrated intensity. Quantifying further and integrating the total intensity together with integrating the north-south and east-west side,

$$I_{\text{tot}} = I_N + I_S, \quad A = \frac{I_N - I_S}{I_N + I_S}.$$

Quantity	McStas	McMini
Total intensity I_{tot}	8.604235×10^1	8.604149×10^1
North–south asymmetry $(N - S)/(N + S)$	3.08×10^{-5}	9.81×10^{-6}
East–west asymmetry	-9.87×10^{-6}	2.48×10^{-5}

Table D.2: Integrated intensity comparison for the flat monitor.

Both implementation agree excellently on the total intensity down to the 10^{-2} , and both implementations exhibit symmetry to within $\mathcal{O}(10^{-5})$, with no systematic north–south or east–west bias. This confirms that the neither the source model or the flat monitor introduces any significant bias. Any asymmetry in later 4π monitor results must therefore originate from angular binning or scattering/recording chain rather than from the source itself.

D.2.2 Spherical monitor

Comparison with McStas 4π monitor

To verify the 4π monitor, again a comparison with an instrument containing only a source and a monitor was made. This time the monitor is a PSD_monitor_4PI with the parameters in Table C.2: The comparison include

Component	Parameters
Source_simple	$\lambda_0 = 4.0$, $\Delta\lambda = 0.1$, focus_xw=focus_yh= 0.5, dist= 1.0, radius= 0.25, $n = 10^9$
PSD_monitor_4PI	radius= 1.0, nx= 3600, ny= 1800, restore_neutron= 1

Table D.3: Simulation parameters for the source and 4PI PSD monitor.

the same metrics used in the flat monitor inspection. Raw data and the difference map can be seen in Figure D.3.

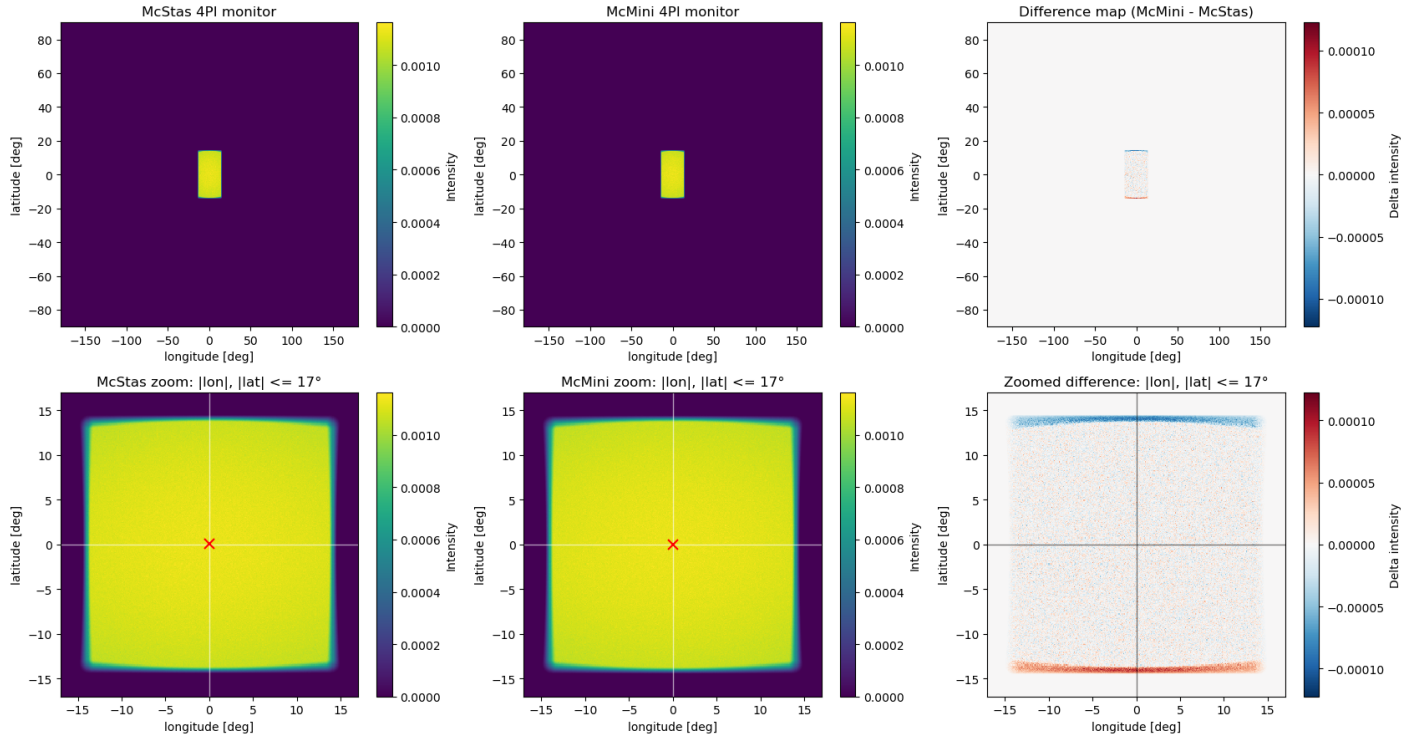


Figure D.3: Comparison, zoomed comparison and difference map of direct beam on a 4PI monitor.

The same color scale had again been forced. Visually they are indistinguishable, however, this time when subtracting the data generated in McStas from that generated in McMini there is something left other than just noise. There is a clear big negative difference and a clear positive difference located at the top and bottom of the data.

4PI monitor symmetry diagnostics

This asymmetry artifact was initially discovered during the multiple scattering simulation with spherical and cylindrical samples. This caused some confusion since it was unexpected that symmetrical sampled would yield asymmetrical results. This problem was investigated and has been narrowed down to a PSD_monitor_4PI issue. Flipping the south hemisphere onto the north hemisphere we see where the greatest discrepancy is. This was done for both McStas and McMini in Figure D.4

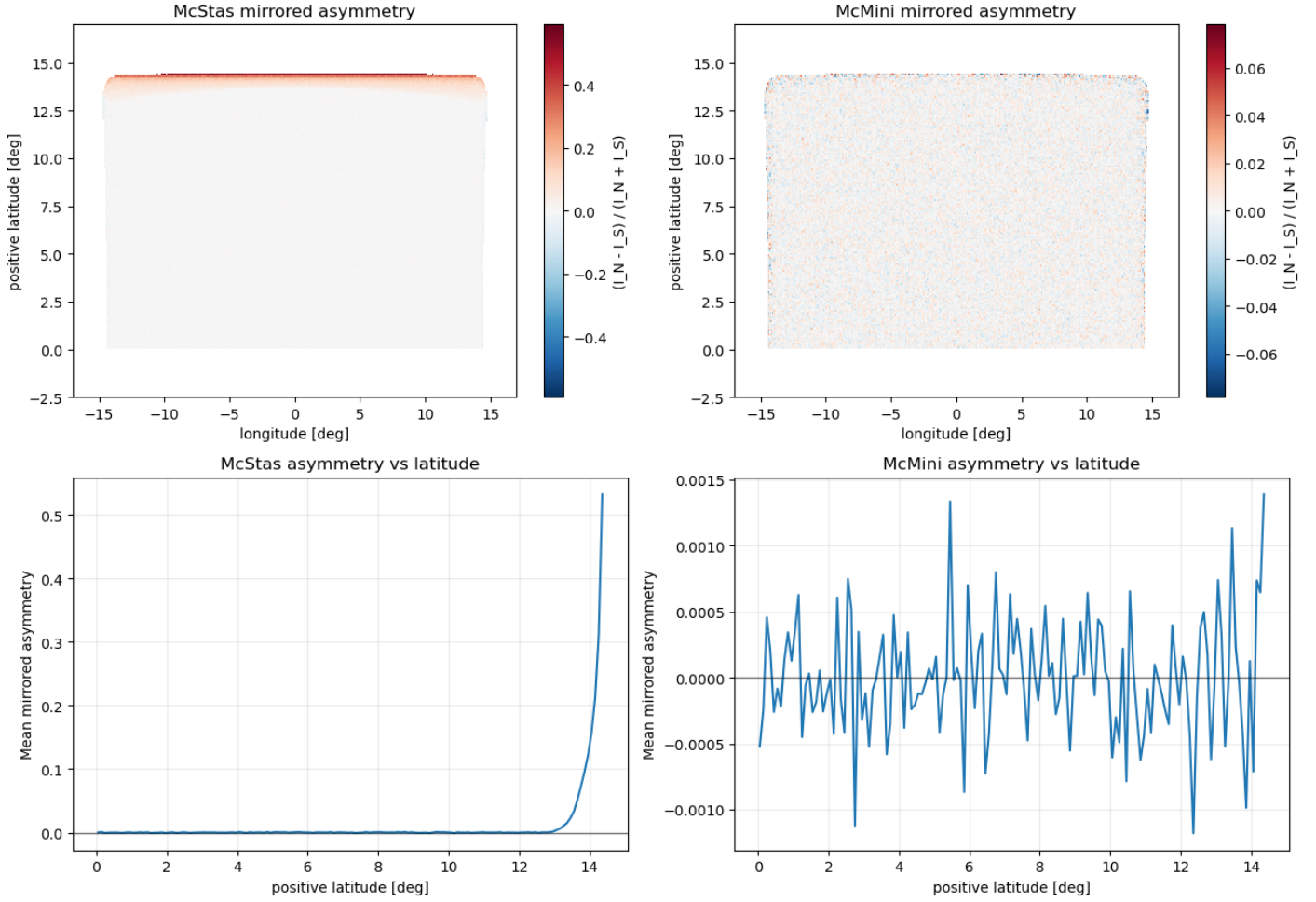


Figure D.4: Mirrored map and asymmetry vs latitude

The figure shows a major asymmetry located at the edge in the McStas simulation. Asymmetry vs latitude shows exponential growth towards the edge at 14° . For McMini the asymmetry is essentially zero, and only noise is left for the mirrored map and asymmetry vs latitude.

Integrated intensity Table D.4 shows the integrated intensity values. It is again seen that the total intensities agree to a very high degree, but McStas shows a north to south asymmetry of 0.37%, whereas McMini agree down to the 10^{-6} .

Quantity	McStas	McMini
Total intensity I_{tot}	8.604208×10^1	8.604149×10^1
North hemisphere N	4.317913×10^1	4.302105×10^1
South hemisphere S	4.286295×10^1	4.302044×10^1
North-south asymmetry $(N - S)/(N + S)$	3.674779×10^{-3}	7.121273×10^{-6}

Table D.4: Integrated intensity comparison for the 4π monitor.